CHSS Data Environment

Hive User Guide

Intake, de-identification, review, preparation and release of CHSS data

For CRQ staff, analysts, researchers and administrators

Follows the protocol CHSS Data Environment Modernization

Version 1.0 · October 2026

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1. About this guide

Hive is the application that runs the CHSS data environment on Cloudera. It takes patient documents in from the sites, removes identifying information, files the documents under each patient's code, and moves de-identified data on to approved research projects. This guide explains how to use every part of Hive, step by step, and how each step follows the protocol **CHSS Data Environment Modernization**.

1.1 Who should read what

You only need the chapters that match your work. Your role decides which pages you see in Hive (chapter 3).

If you are…	Read these chapters
Everyone	1 About this guide, 2 The CHSS data environment, 3 Roles and access, 4 Getting started
CRQ intake staff	5 Zone 1: intake and review, 6.1 Consent matrix
Senior data analyst	5.7–5.8 Files and Metadata, 6 Zone 1X: preparation
Researcher or analyst on a project	7 Zone 2: release and analysis, 6.3 Projects (Members and Safe copy)
CHSS Manager	All chapters; especially 5.6 Originals, 7.1 Zone 2 inbox, 9 Administration
CHSS Application Specialist	All chapters; especially 9 Administration and 10 Reference

1.2 How to read the guide

- Names of buttons, fields and pages are written in **bold**, exactly as they appear on screen.
- Numbered lists are steps to follow in order.
- Protocol boxes quote the rule from the protocol that a step follows, with its section number (for example §9.1.2).
- Boxes marked **Not yet available in Hive** describe parts of the protocol that are not built yet. They explain how the step will work, with a placeholder where its picture will go.

> **Note**
> All patients, documents, users and projects in the screenshots are made up for this guide (for example patient AVSD0136 "Liam Practice"). No real patient data is shown.

2. The CHSS data environment

2.1 Zones

The protocol splits the environment into zones. Each zone has its own purpose, its own users and its own rule on identifying information (PHI). Data only moves forward, from Zone 1 to Zone 2, and becomes less identifying at every step.

Zone	Purpose	Who can use it	PHI	In Hive
Zone 1	Data intake, integration and storage	CRQ staff	Yes	Sidebar group **Zone 1 · Intake & review**
Zone 1X	Data cleanup and preparation, one area per project (1A, 1B, …)	Approved project staff	Minimal or incidental only	Sidebar group **Zone 1X · Preparation**
Zone 2	Analysis of an approved study	Approved members of that study	None	Sidebar group **Zone 2 · Release**, and the project's Zone 2 folder
Zone 3	Document repository (SharePoint)	CHSS team	None	Outside Hive
Zone 4	Internet: REDCap, site uploads, patient portal	Sites, patients	No identifiers	Outside Hive (REDCap exchange in Zone 1)

[Picture to be added]

Copy Figure 1 of the protocol here: site upload → Zone 1 storage → Zone 1X cleaning → temporary storage → analysis preparation → Zone 2 storage.

Figure 1. The zones of the CHSS data environment (protocol Figure 1)

2.2 How data moves

1. **Sites send files.** Sites upload each patient's documents (PDFs, images, forms). Files arrive in the intake folder (chapter 5.1).
2. **De-identification.** Hive reads each file, covers identifying text with black boxes, gives it a standard name, and files it under the patient's code (chapter 5.2).
3. **Review.** CRQ staff check each patient's documents against the four-point enrolment checklist and submit or reject the application (chapter 5.3).

4. **Preparation (Zone 1X).** For an approved project, analysts pick the patients, build a workspace and clean the data (chapter 6).
5. **Safe copy.** Hive removes the remaining PHI, shifts dates, gives every patient a project-only ID and places the safe copy in the Zone 2 inbox (chapter 6.3).
6. **Release (Zone 2).** The CHSS Manager checks the safe copy and releases it to the project's Zone 2 folder, where the researchers work (chapter 7).

2.3 Patient codes (UBRS)

Every patient is known by a code, never by name, outside Zone 1. In Hive the code is read from the file's folder or file name. The accepted format is shown at the top of the De-Identifier page: two to four letters followed by three or four digits, for example **AA0001** or **AVSD0001**. Files without a valid code are not de-identified; they are listed as **Excluded** (chapter 5.2).

> **Important**
> A typing mistake that still looks like a valid code (AVSD0137 instead of AVSD0136) cannot be caught by Hive. Always check that the code shown on screen is the right patient.

2.4 Document names

Every de-identified file gets a standard name made of the patient code, the document type, the document date and a unique 16-digit serial number, followed by the redaction mark:

`AVSD0136_ECHO_REPORT_20190520_1793379635294000_R.pdf`

Part	Example	Meaning
Patient code	AVSD0136	The UBRS code of the patient
Document type	ECHO_REPORT	Enrolment form, contact form, consent, operative report, discharge summary, echo, angiography, MRI, ECG, image, STS extract or other. Images use their DICOM modality (US, MR, CT…). UNKNOWN if not found.
Document date	20190520	Date of the document (YYYYMMDD), or UNKNOWN
Serial	1793379635294000	16-digit number, unique for every document
Redaction mark	_R	**_R** = redacted automatically by Hive. **_RR** = redacted again by hand (chapter 5.5).

> **Protocol**
> §9.1.2 and §9.1.4: files are renamed with the UBRS prefix, document type, document date and a 16-digit serial number; the automatically redacted file gets the suffix "_R", and a file redacted again by hand gets an extra "R".

3. Roles and access

3.1 Getting access

Access to each zone is given only on approval, need and completed training.

- **Training.** Before access is given you must complete TCPS2 and the iLearn privacy training (or the US equivalent). The CHSS Manager keeps the proof.
- **Sign-in.** Hive has no password of its own. You open it from Cloudera, which signs you in with your hospital account (Entra ID with multi-factor authentication). Zones 1 and 1X also need the hospital VPN.
- **Account.** The CHSS Manager or Application Specialist creates your Hive account and gives it a role (chapter 9.1). If you can open Cloudera but Hive says you have no account, ask for one.
- **Review.** The CHSS Manager reviews who has access. Zone 2 access is limited to one project and ends when the project ends; it is reviewed every 6 months.

> **Not yet available in Hive**
>
> §5: Zone 2 access is revoked automatically every 6 months unless it is extended.
>
> In Hive today, the CHSS Manager removes access by hand: remove the person from the project's **Members** (chapter 6.3) or change their role on the **Users** page.

3.2 Roles

A role is a set of permissions. The table shows the roles used in this guide. The CHSS Manager can rename them, change what each one may do, or add new ones on the **Roles** page (chapter 9.2).

Role	Typical person	What the role does in Hive
CHSS Manager	Program manager	Everything. Approves access, grants access to archived originals, reviews the Zone 2 inbox and releases data, manages security incidents.
CHSS Application Specialist	Runs the platform	Everything except granting originals and releasing to Zone 2: users and roles, De-Identifier runs, REDCap exchange, projects and workspaces, security monitoring.
CRQ Intake Staff	Intake and follow-up staff	Zone 1: copies files in, runs the De-Identifier, reviews applications and the enrolment checklist, handles rejections, patients, REDCap exchange and the consent matrix. Gets the intake emails.
Senior Data Analyst	Senior analysts, biostatisticians	Reads de-identified Zone 1 documents and metadata; Zone 1X: abstraction folders, projects, cohorts, workspaces and preparing the safe copy.

Role	Typical person	What the role does in Hive
Researcher	Analysts on an approved study	Sees the projects they are a member of and their own abstraction folders. Works on the released data in the project's Zone 2 folder.
Viewer (read-only)	Auditor	Can look at patients, applications, files and logs but change nothing.

What you see depends on your role. Pages you have no permission for are not in your sidebar. The pictures below show the same Dashboard for two different roles.

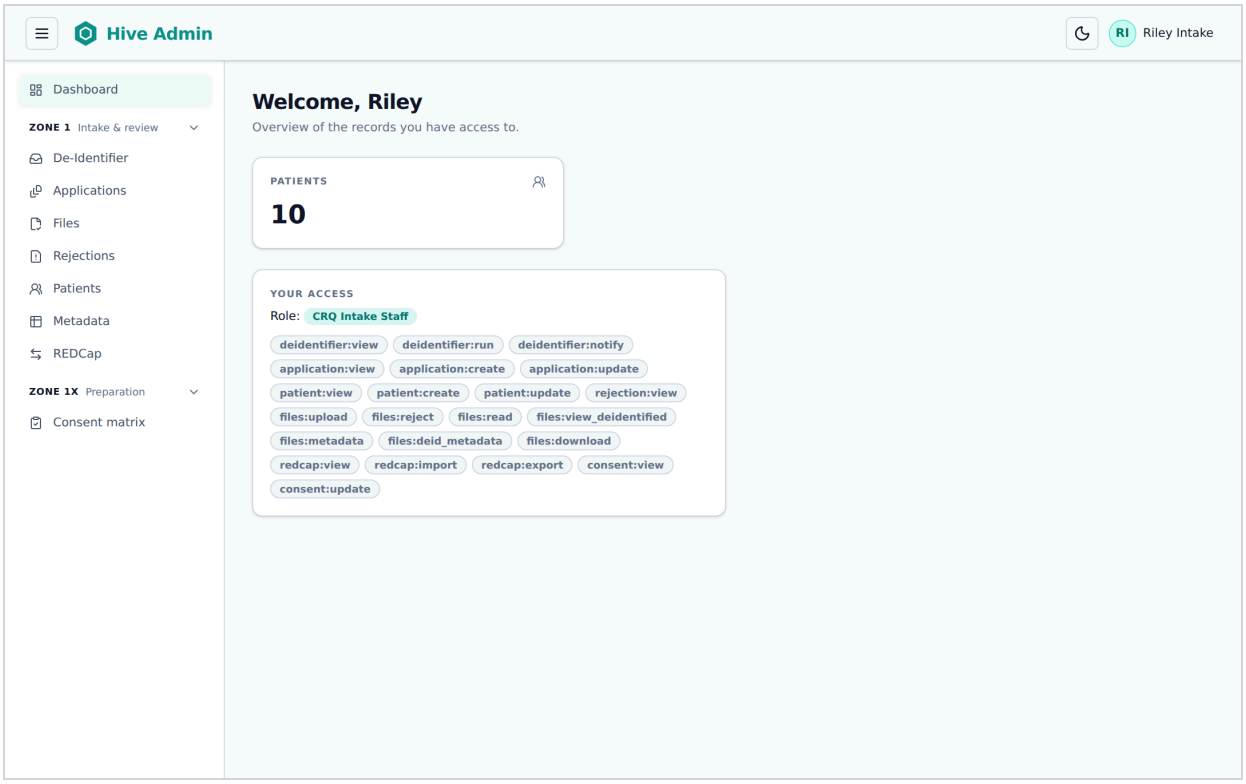


Figure 2. Dashboard for CRQ Intake Staff: Zone 1 pages and the consent matrix

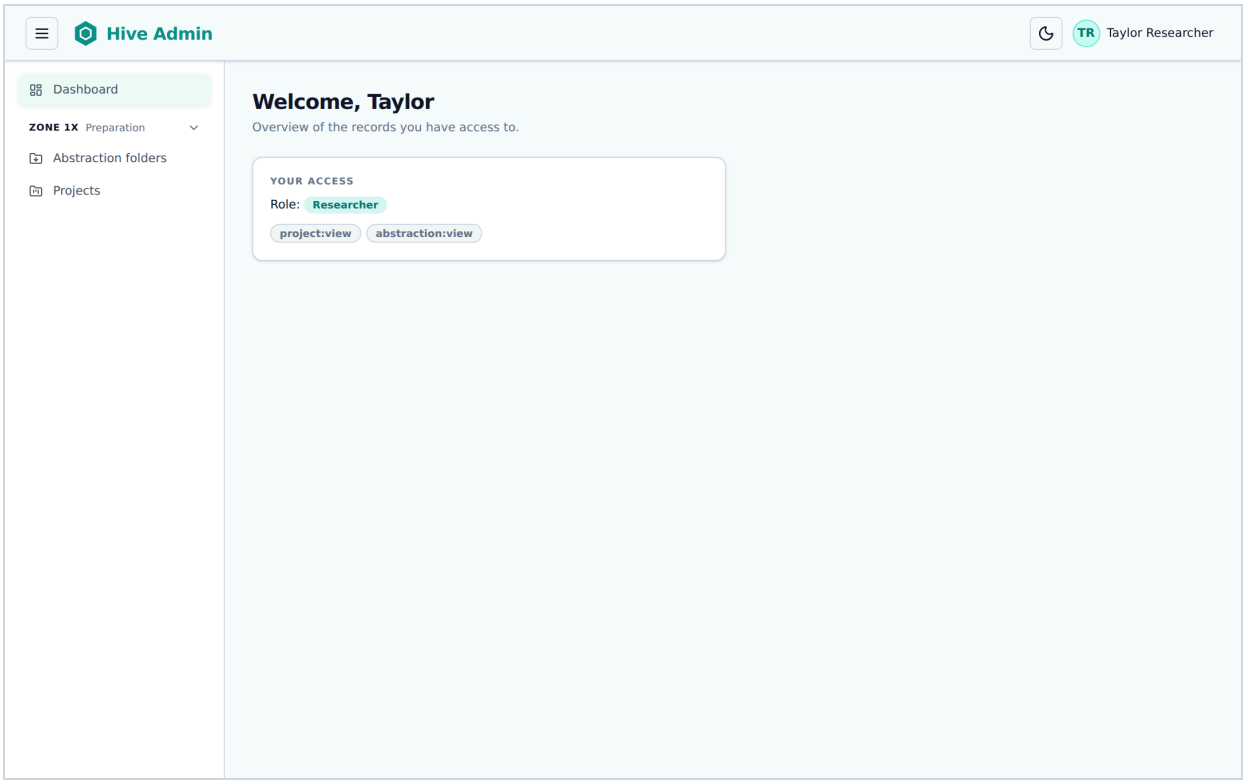


Figure 3. Dashboard for a Researcher: only Abstraction folders and Projects

4. Getting started

4.1 Opening Hive

1. Connect to the hospital VPN (needed for Zones 1 and 1X).
2. Open Cloudera and sign in with your hospital account and multi-factor authentication.
3. Open the **Hive** application from the CHSS project. Hive opens on the **Dashboard**.

4.2 The screen

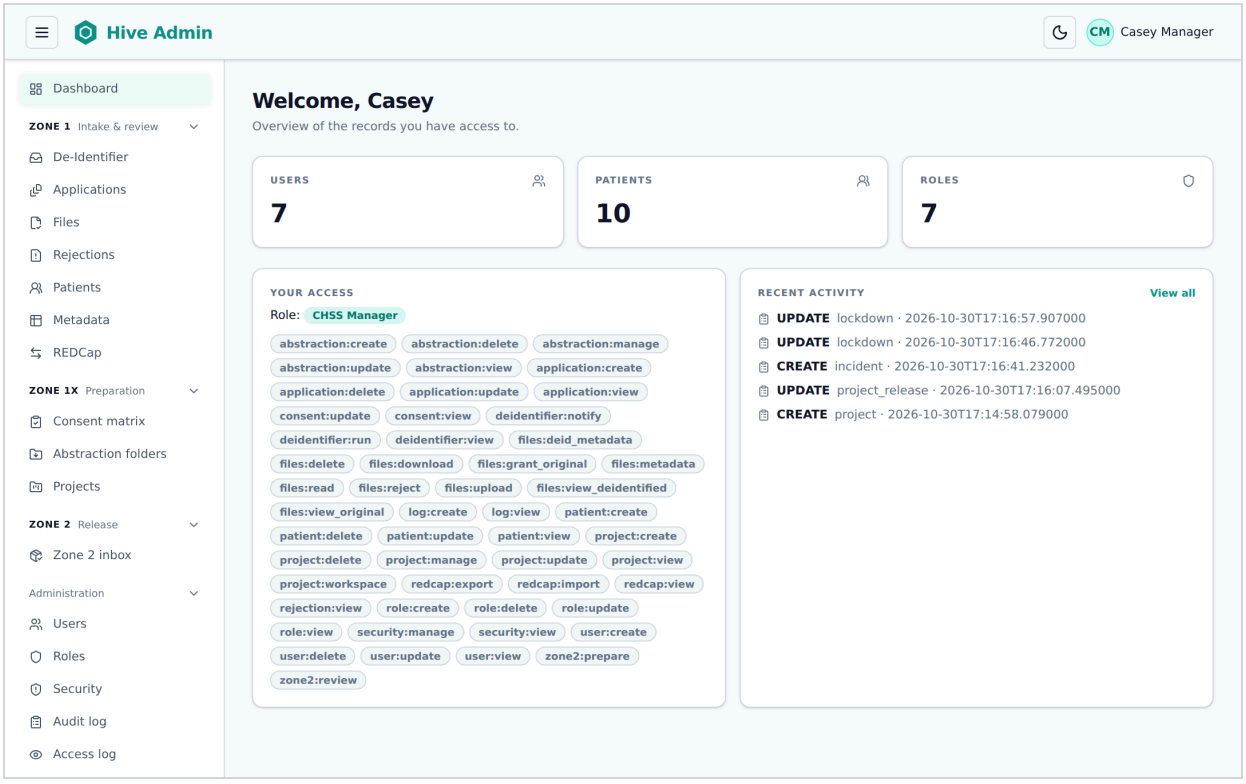


Figure 4. The Dashboard for the CHSS Manager

- **Sidebar (left).** Pages grouped by zone: **Zone 1 · Intake & review**, **Zone 1X · Preparation**, **Zone 2 · Release** and **Administration**. Click a group name to fold it. The ☰ button at the top hides the sidebar.
- **Top bar (right).** The moon button switches between light and dark colours. Your name opens **Your Profile**, which shows your role and permissions.
- **Zone badge.** Every page title has a coloured badge (for example **Zone 1 · Intake & review**) so you always know which zone you are working in.
- **Dashboard.** Counts of users, patients and roles, your access, and the most recent changes.

4.3 Tables

Most pages show a table. You can search with the box above it, sort by clicking a column name, choose how many **Rows** to show and move between pages at the bottom. The ⋯ button at the end of a row opens the actions for that row.

4.4 When Hive is locked

During a security incident the CHSS Manager can lock Hive (chapter 9.3). A red banner then appears on every page and nothing can be opened or changed. Wait until it is unlocked; do not try to work around it.

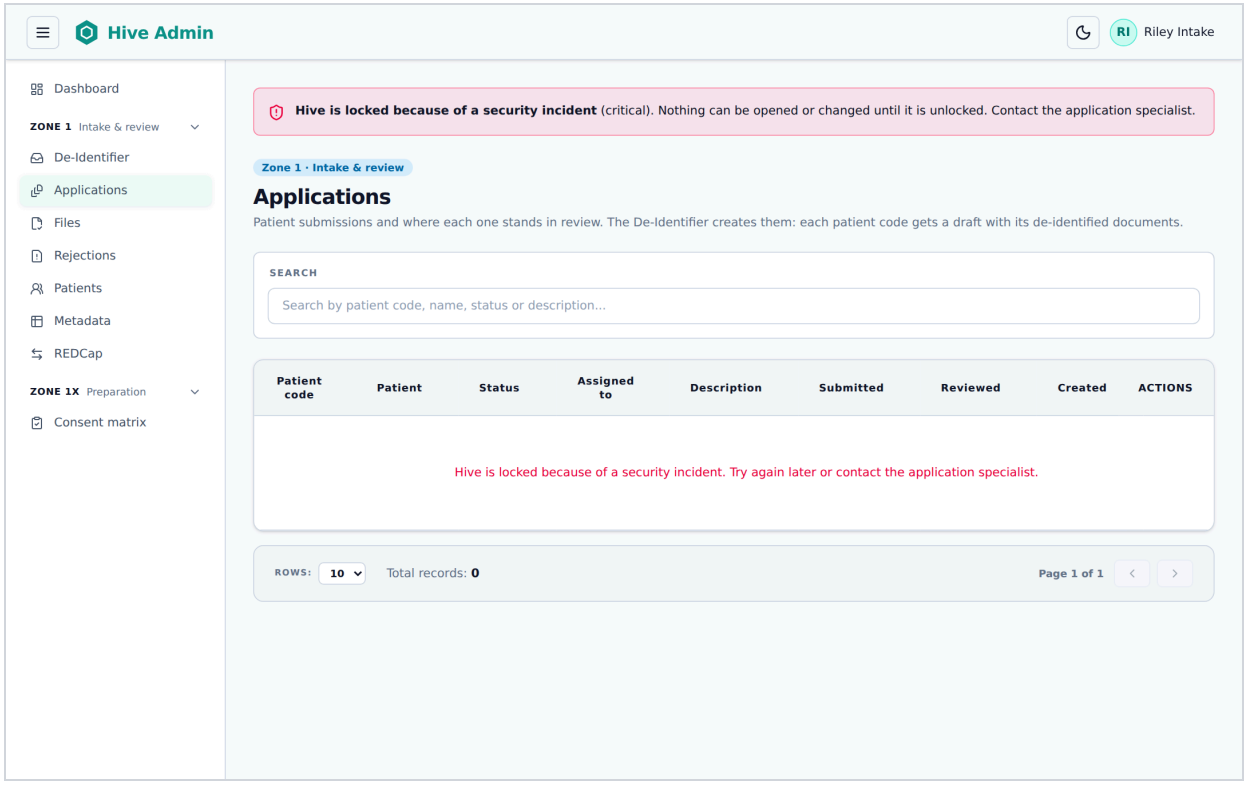


Figure 5. What staff see while Hive is locked

5. Zone 1: intake and review

Zone 1 is where identified data is received and turned into de-identified, named and checked documents. Only CRQ staff work here.

5.1 How files arrive

From the sites: the SFTP inbox

Sites send each patient's files securely by SFTP. Files land in the intake folder (the inbox). Only the Hive application reads this folder; no person opens it.

> **Protocol**
>
> §9.1.1: Each upload (transaction) is one patient. Incoming files are scanned for malware and checked for integrity. Files of a known patient carry the UBRS as the file name prefix.
>
> If a file fails the scan or has a missing or wrong UBRS, it is deleted and the person who sent it is emailed. CRQ staff are emailed when new data has been processed.

> **Not yet available in Hive**
>
> The SFTP connection for sites and the malware scan are set up outside Hive and are not running yet.
>
> Today: files with a missing or wrong code are not deleted; Hive lists them as **Excluded** on the De-Identifier page (chapter 5.2) so the site can send a corrected file. Hive emails CRQ staff (not the site) when a run finishes.

[Picture to be added]

Picture of the site's SFTP upload screen and the inbox folder, once the SFTP connection is in place.

Figure 6. SFTP upload by a site

From the Carbon share: Copy from Carbon

Historical documents and images kept on the Carbon research share are brought in with **Copy from Carbon** on the De-Identifier page. Hive copies the folders you pick into the intake folder; the files on Carbon are not changed.

1. Open **De-Identifier** in the sidebar.
2. In the **Copy from Carbon** card, click **Browse Carbon**.

3. Click a folder name to open it, or use the path at the top to go back. Tick the folders or files you want. **Select everything in this folder** ticks all of them.
4. Click **Check**. Hive counts the new files and their size. Files already in the intake folder or already processed are skipped and counted separately.
5. Click **Copy N files**. A progress bar shows the copy. You can leave the page; the copy continues.
6. When the copy is done, start de-identification (chapter 5.2).

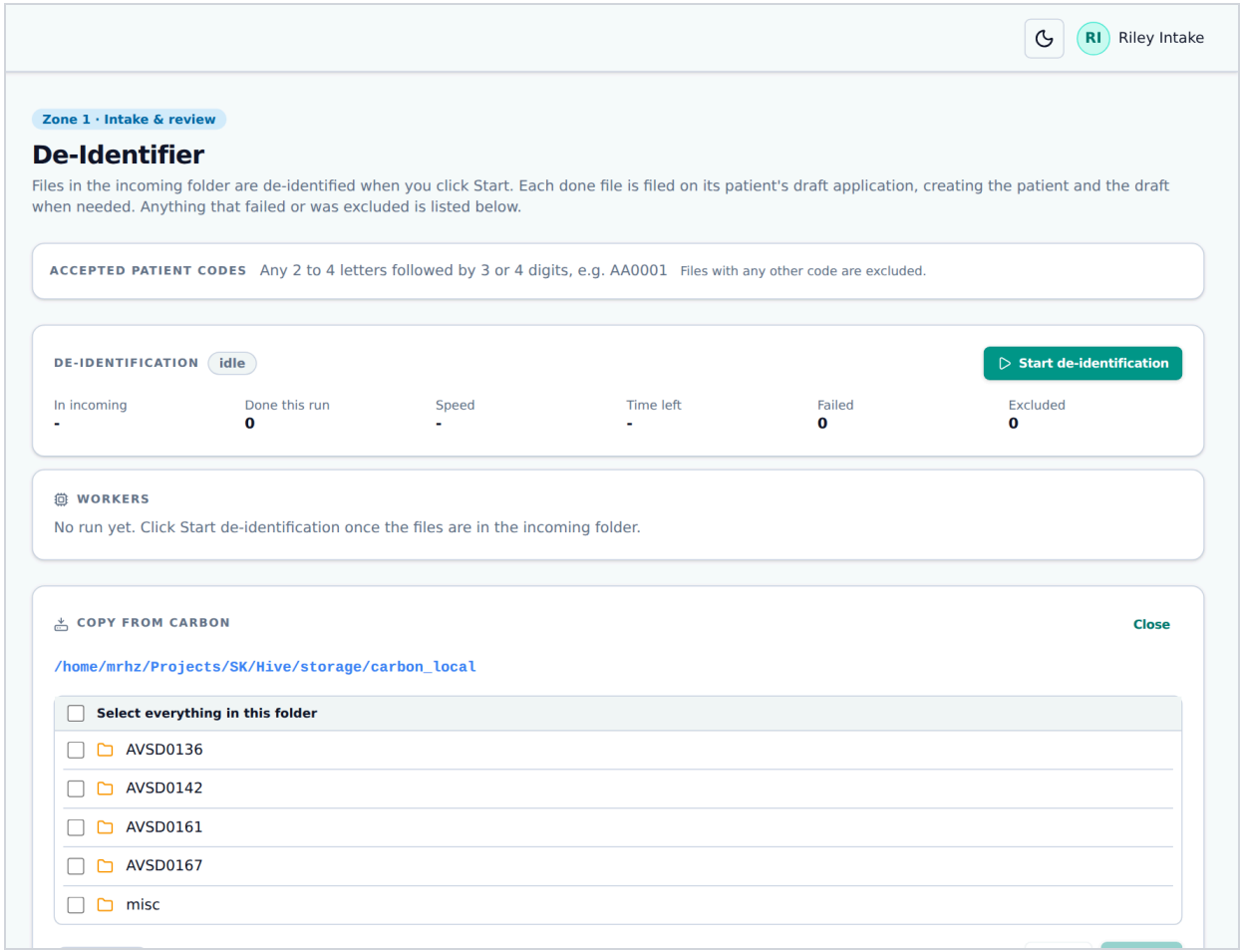


Figure 7. Browsing the Carbon share

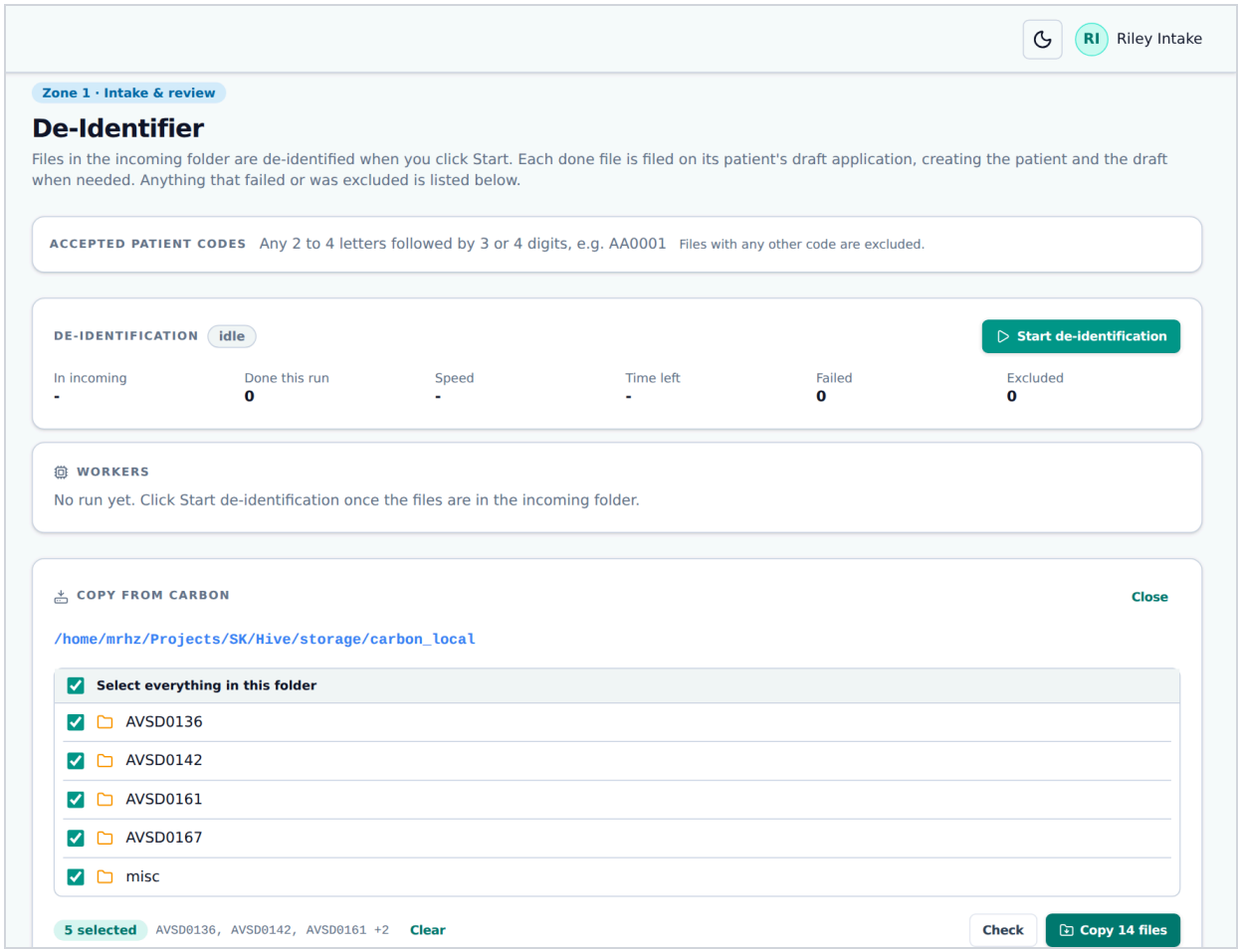


Figure 8. After Check: 14 new files are ready to copy

5.2 De-Identifier

The **De-Identifier** page de-identifies everything waiting in the intake folder. Each finished file is filed on its patient's **draft application**. If the patient does not exist yet, Hive creates the patient (code only) and the draft application.

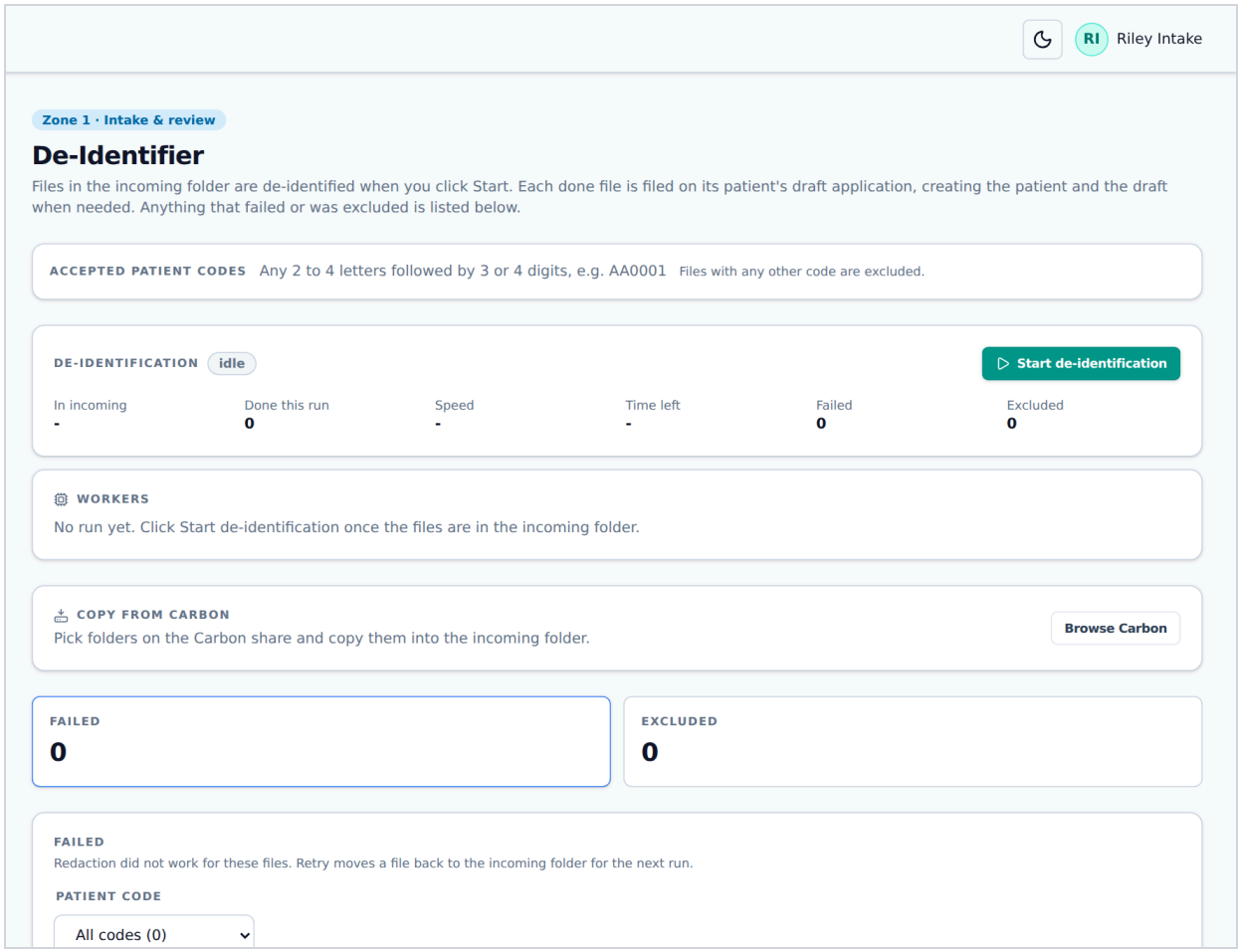


Figure 9. The De-Identifier page

Running de-identification

1. Check **Accepted patient codes** at the top. Files with any other code are excluded.
2. Click **Start de-identification**. The status changes to **running**.
3. Follow the progress: **In incoming** (still waiting), **Done this run**, **Speed**, **Time left**, **Failed** and **Excluded**. **Workers** shows the machines doing the work, and **Finished this run** lists each file as it is done.
4. When the status is back to **idle**, the run is finished. Staff who get intake emails receive an **Intake run finished** email with the counts.

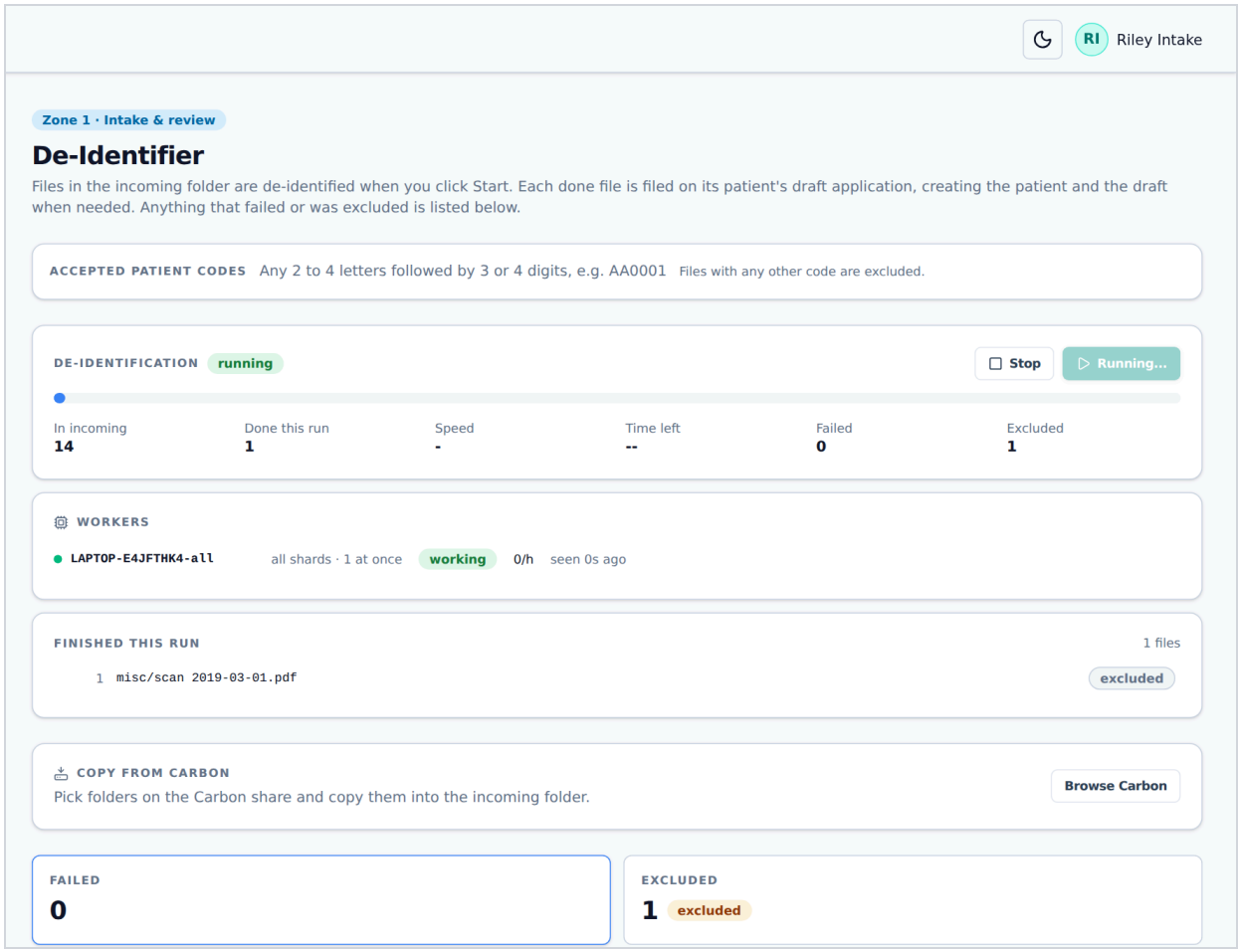


Figure 10. A run in progress: one file done, one excluded

> **Tip**
>
> You can click **Stop** at any time. The file being worked on is done again in the next run; files already finished are kept. Click **Start de-identification** again to continue.

What de-identification does

- **PDF and Word documents.** Text is read from every page with PaddleOCR, including scanned pages. Names, dates, record numbers, addresses, phone numbers and other identifiers are found with Presidio (spaCy and the Stanford de-identifier model) and covered with black boxes using MuPDF. The text under each box is removed, not just hidden. The document's properties (author, title) are cleaned too.
- **DICOM images.** Identifying header fields are removed or blanked, private tags are removed, the image IDs (UIDs) are replaced, and text burned into the picture is covered.
- **JPG, PNG, TIFF and other pictures.** They are first converted to DICOM with the patient code, modality and date, then treated like DICOM images.
- **Name.** The copy gets its standard name ending in **_R** (chapter 2.4). The original is kept in the archive (chapter 5.6).

Protocol

§9.1.4: all PDFs go through automated redaction. Python with PaddleOCR finds typed and handwritten text with its location; Presidio with spaCy en_core_web_sm and StanfordAIMI stanford-deidentifier-base finds PHI; MuPDF redacts it. The redacted PDF, with the suffix “_R”, is the approved version for abstraction and analysis. §9.1.8: non-DICOM images are converted to DICOM.

Failed and Excluded files

The two cards under the progress show files that need attention. Click a card to see its list.

List	Why a file is here	What to do
Failed	Redaction did not work (for example a damaged file, or the run was stopped by a fault).	Click **Retry** on the row, or **Retry all**. The file goes back to the intake folder for the next run.
Excluded	No patient code; a format Hive cannot read; an exact copy of a file already processed; or the folder and the file name have different codes.	For a code conflict, click the right code on the row (it shows whether the code came from the **path** or the **name**). For everything else, ask the site to fix and resend the file, then click **Retry**.

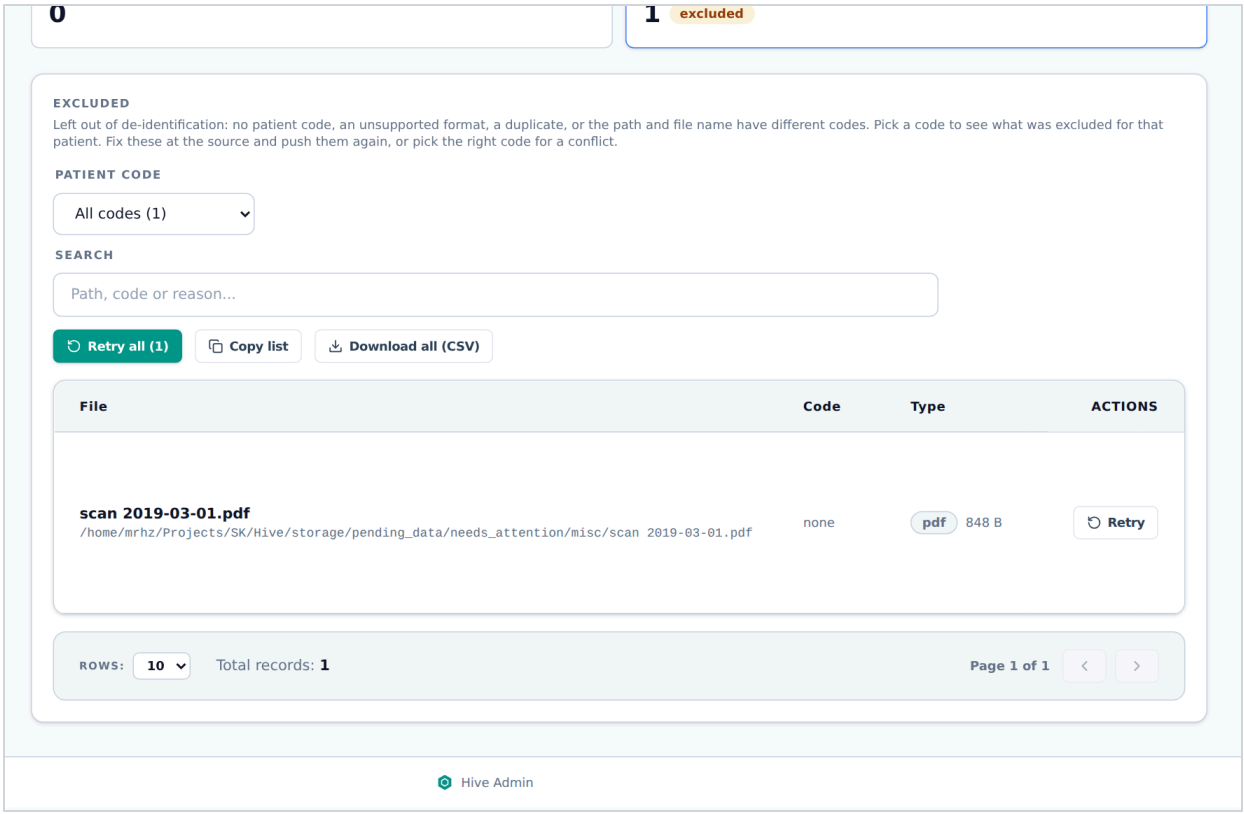


Figure 11. An excluded file with no patient code

Copy list copies the list to the clipboard (full path, reason and detail), and **Download all (CSV)** saves every file in the list, not just the ones shown.

5.3 Applications

An application holds one patient's submission: the patient's details and their documents. The De-Identifier creates a **draft** application for each patient code. CRQ staff then review it and either submit or reject it.

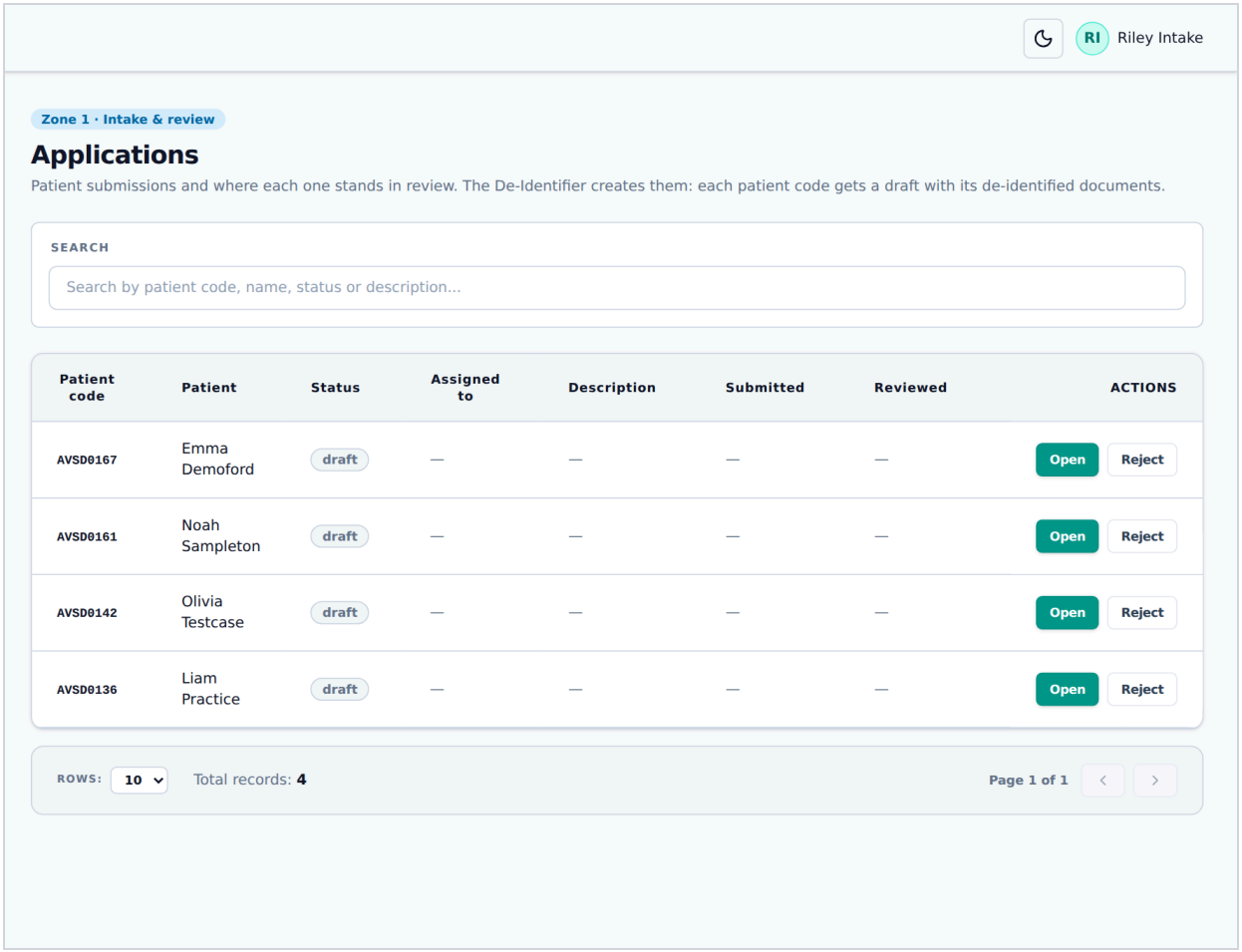


Figure 12. The Applications page

Status	Meaning
draft	Created by the De-Identifier; waiting for review
submitted	Checked and accepted. Cannot be changed any more.
rejected	Turned down with a reason. Listed on the Rejections page until it is fixed or deleted.
deleted	Its documents were deleted (for example patient not eligible). Only the record and reason remain.

Click **Open** on a row to review it. An application has three steps shown at the top: **1. Patient**, **2. Documents** and **3. Summary**.

Step 1: Patient

1. Under **Assignment**, pick who is reviewing this application in **Assigned to**. That person is emailed about uploads to it.
2. Check the patient's details. If the patient came from the bulk import or the old data, a yellow box says **Patient details not checked yet**. Correct anything wrong, then click **Details are correct**. This is asked only once per patient.
3. If the site does not share records or contact details, tick **Site does not share records or contact details**. The patient then needs no name and is not held for the details check.
4. Click **Save and continue**.

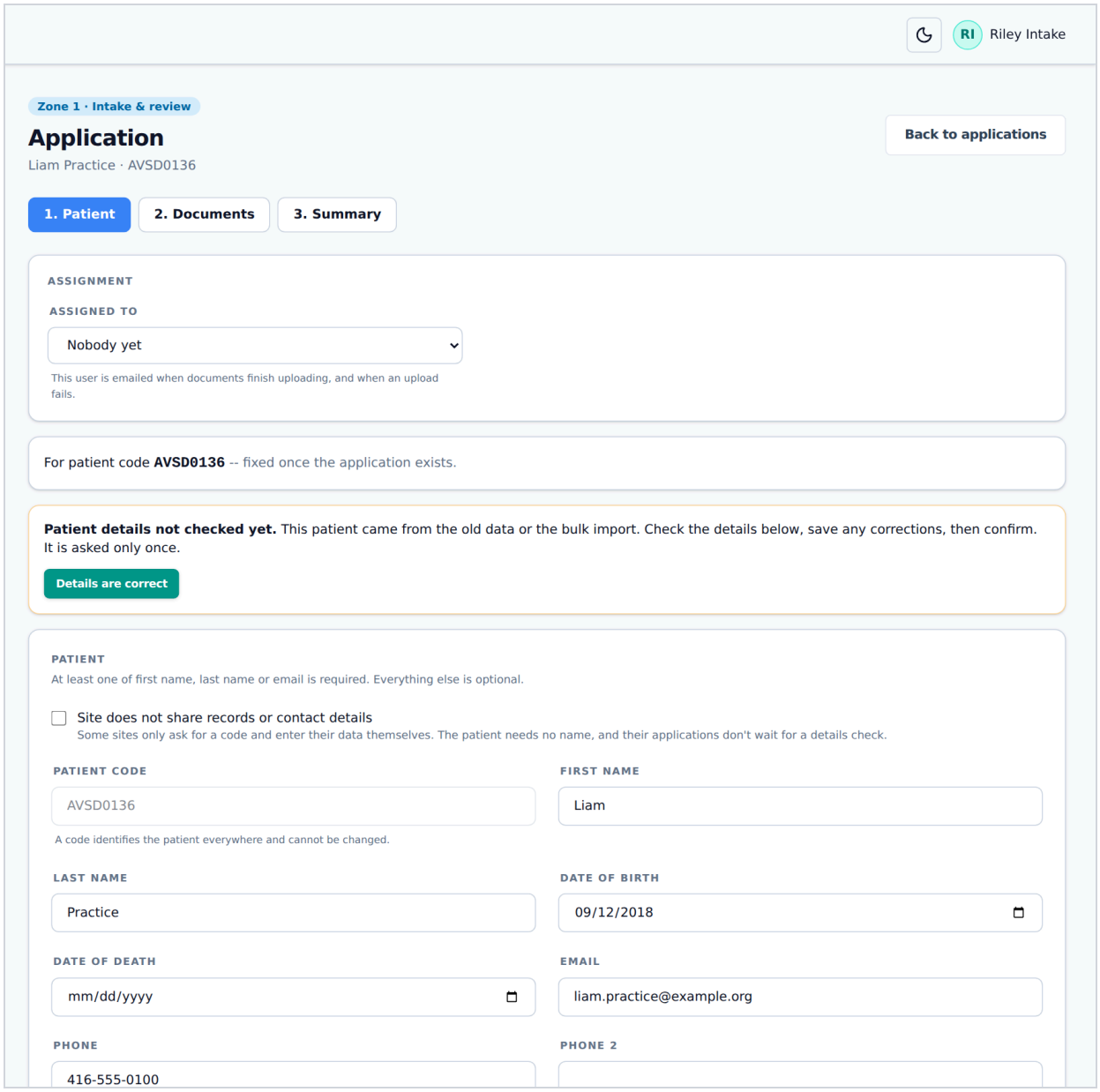


Figure 13. Step 1: checking the patient's details

> **Protocol**
>
> §9.1.2: for an existing patient, check for a change-of-contact form and update the information; for a new patient, record the contact information. Sites that do not share PHI use the "no records / no ID" option and get no missing-information queries.

Step 2: Documents

This step lists the patient's documents with their type and date, review status and de-identification status. The bar at the top counts documents, how many are de-identified and how many are approved.

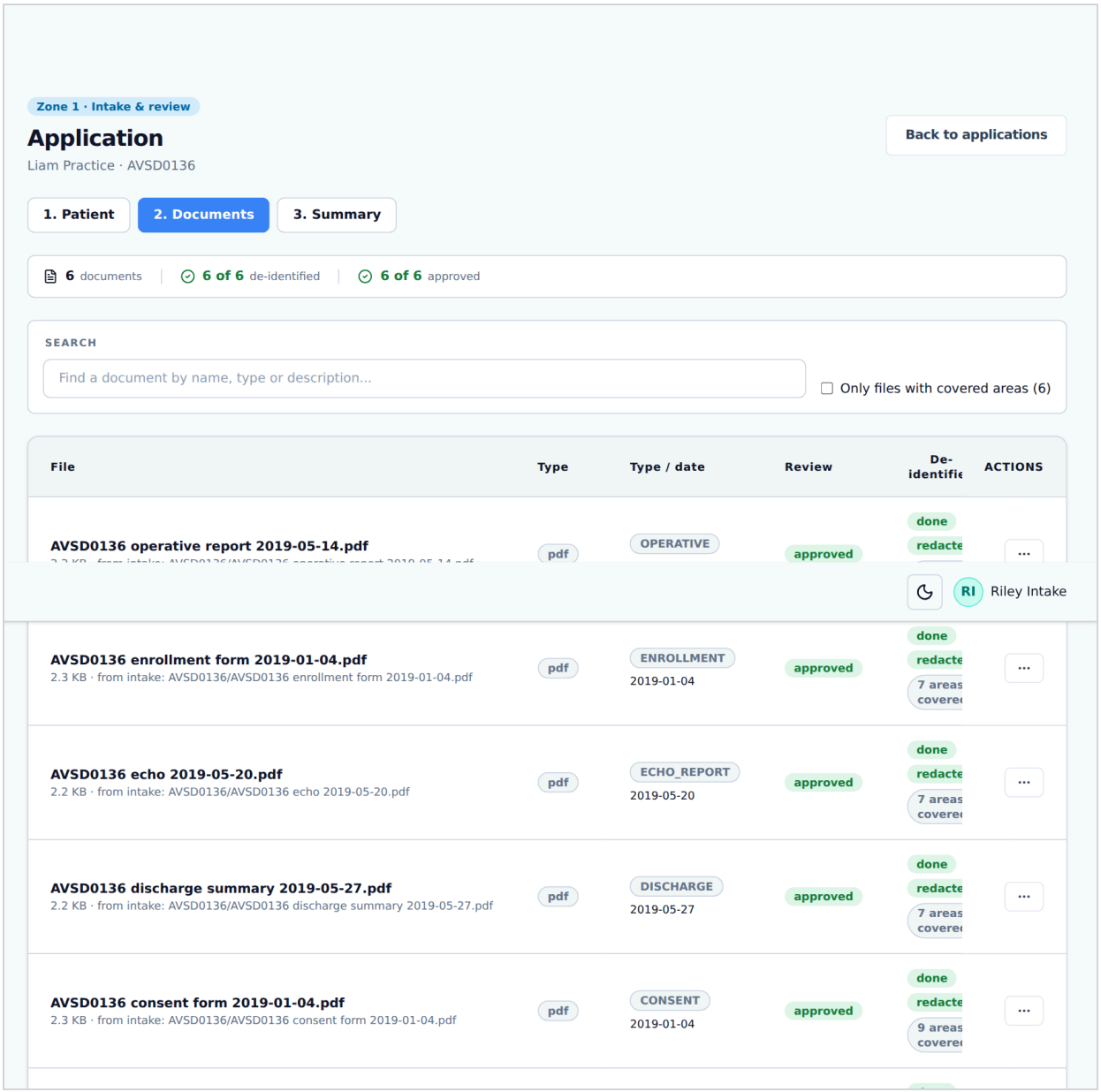


Figure 14. Step 2: the documents of an application

- **Type / date** shows what Hive read from the file name. **type?** or **date?** in yellow means it was not found.
- **De-identified** shows **redacted**, **by hand** (an _RR copy) and how many areas were covered. Tick **Only files with covered areas** to see only those.
- **leaded · not for Zone 2** in red marks an image with identifiers in lead letters (chapter 5.9).

Click ⋯ on a document to see its actions. Which ones you see depends on your role.

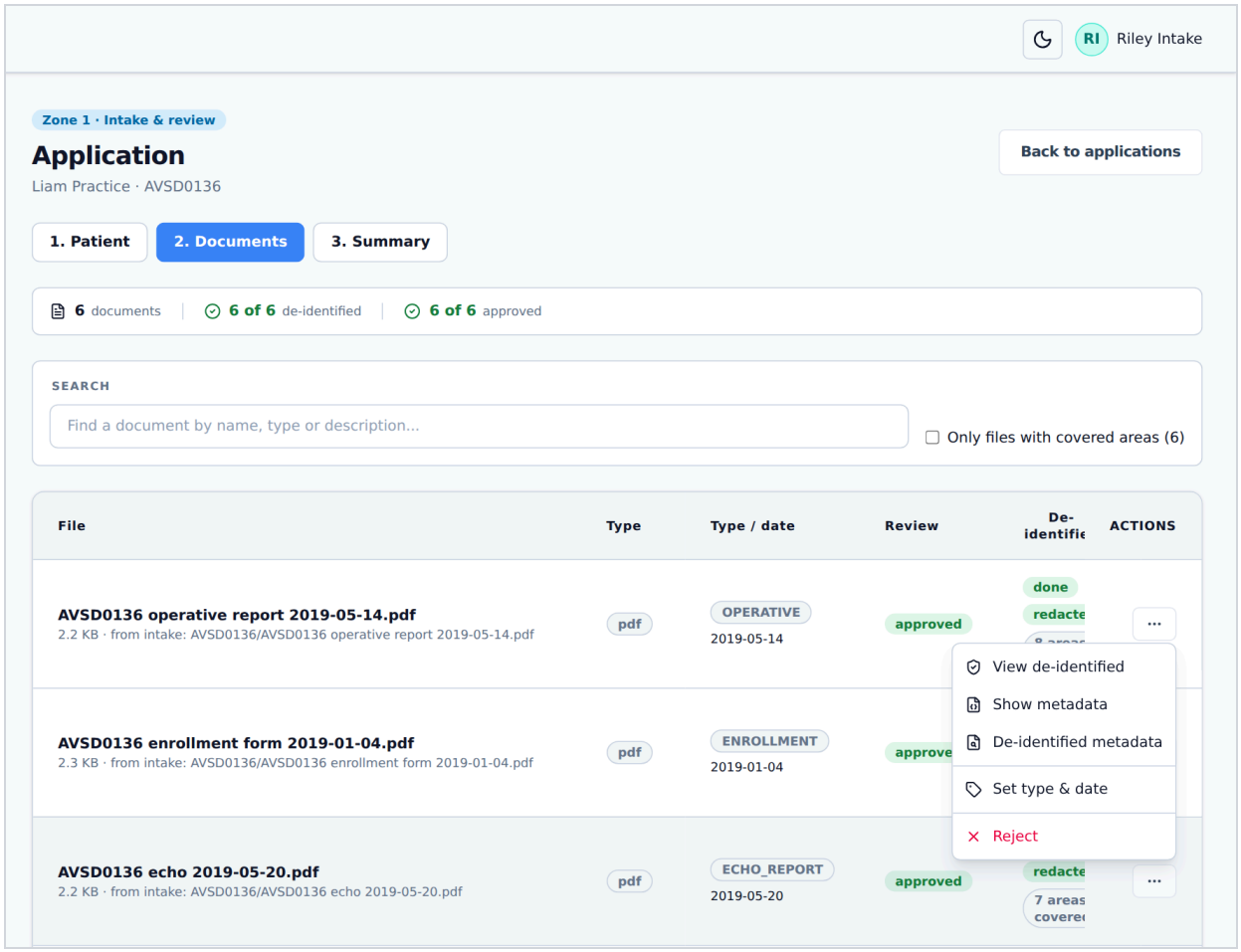


Figure 15. The actions for a document

Action	What it does
View de-identified	Opens the redacted copy. Use it to check that no identifier is left.
View original	Opens the original (only roles allowed to see originals, and only before submission; afterwards see chapter 5.6).
Show metadata / De-identified metadata	Shows the information stored inside the file, before and after de-identification.
Set type & date	Corrects the document type and date. The copy is renamed to match.
Mark as leaded	For images only (chapter 5.9).
Reject	Turns the document down with a reason (for example an identifier is still readable). It goes to Rejections.



Figure 16. Checking the de-identified copy: identifiers are covered with black boxes

Correcting the type and date

1. Click ⋯ → **Set type & date**.
2. Pick the **Document type / modality**. Reports and forms get a document type (for example **Echocardiogram report (ECHO_REPORT)**); DICOM images their modality.
3. Enter the **Document date**, or tick **Date unknown**.
4. Click **Save**. The de-identified copy is renamed.

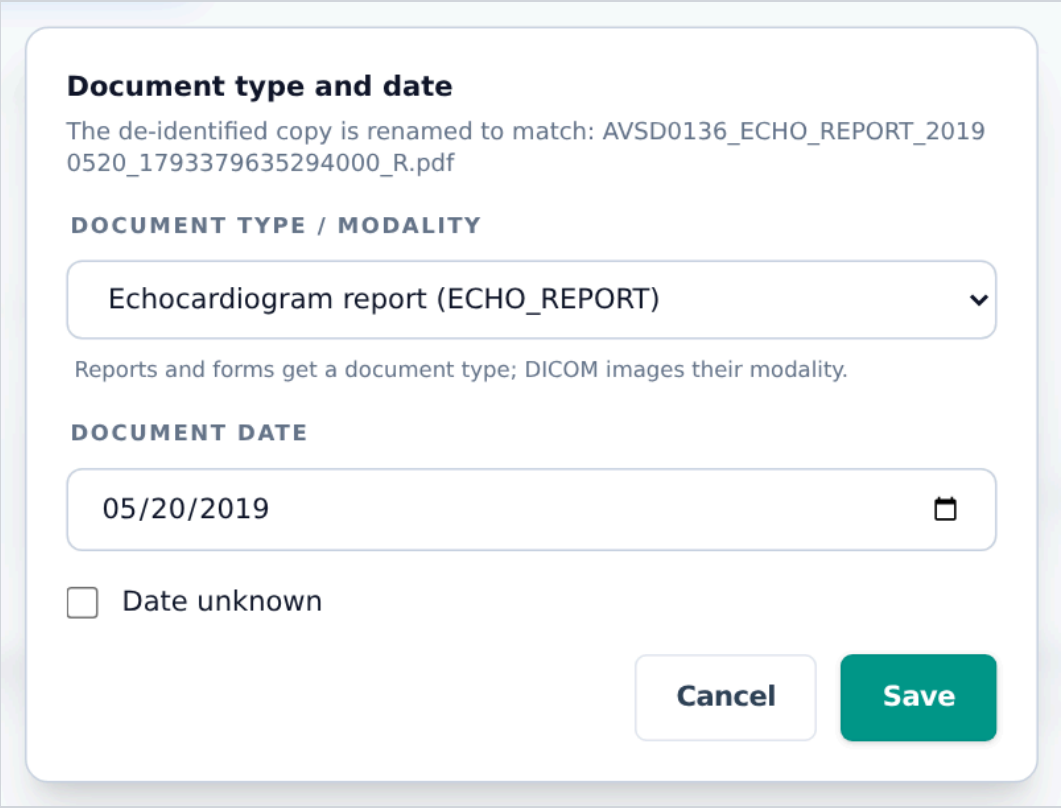


Figure 17. Setting the document type and date

> **Protocol**
> §9.1.2: Python extracts the document type and date; staff validate and correct them, and verify that no identifiers remain.

Step 3: Summary, checklist and submit

The summary shows the patient and every document. **Submit application** only works when the patient's details are confirmed and every document is approved (none undecided or rejected).

1. Click **3. Summary**, then **Submit application**.
2. The **Enrollment checklist** opens. Tick each point only if it is true: **All required files were submitted**, **The patient is eligible**, **The right consent form is present**, **No identifiers were found**.
3. Click **Submit application**. The status becomes **submitted**, the patient code is stamped on every page of the redacted PDFs, and the documents move to the archive.

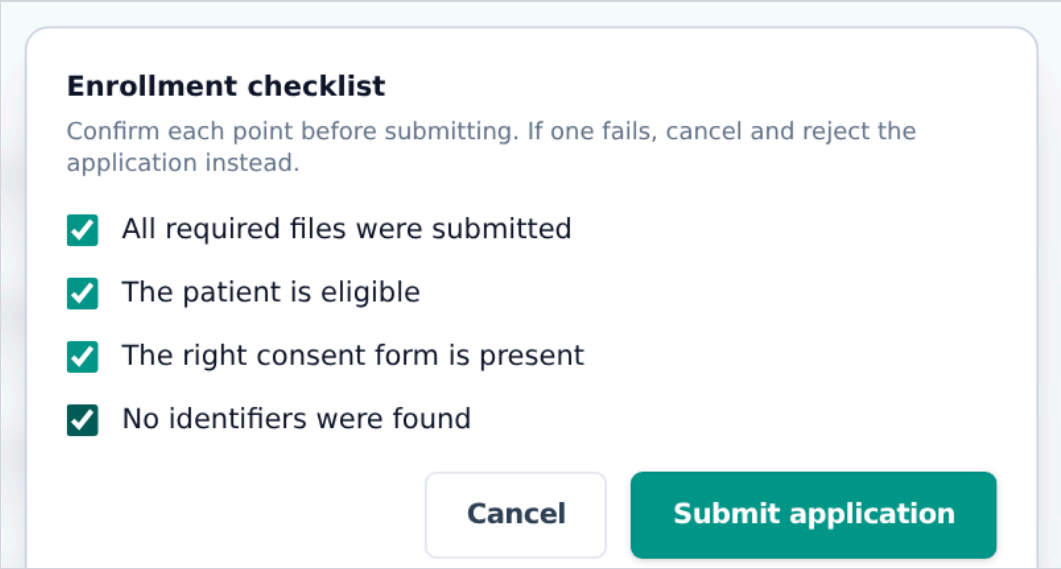


Figure 18. The four-point enrolment checklist

> **Protocol**
> §9.1.2: before a patient is enrolled, staff confirm that (1) all required files were submitted, (2) the patient is eligible, (3) the proper consent forms are present and (4) no identifiers are present.

Rejecting an application

If any checklist point fails, do not submit. Reject the application instead.

1. In **3. Summary**, click **Reject application** (or **Reject** on the Applications list).
2. Under **What failed?**, tick every point that failed.
3. Write the **Reason** so whoever fixes it knows what is wrong.
4. Click **Reject application**.

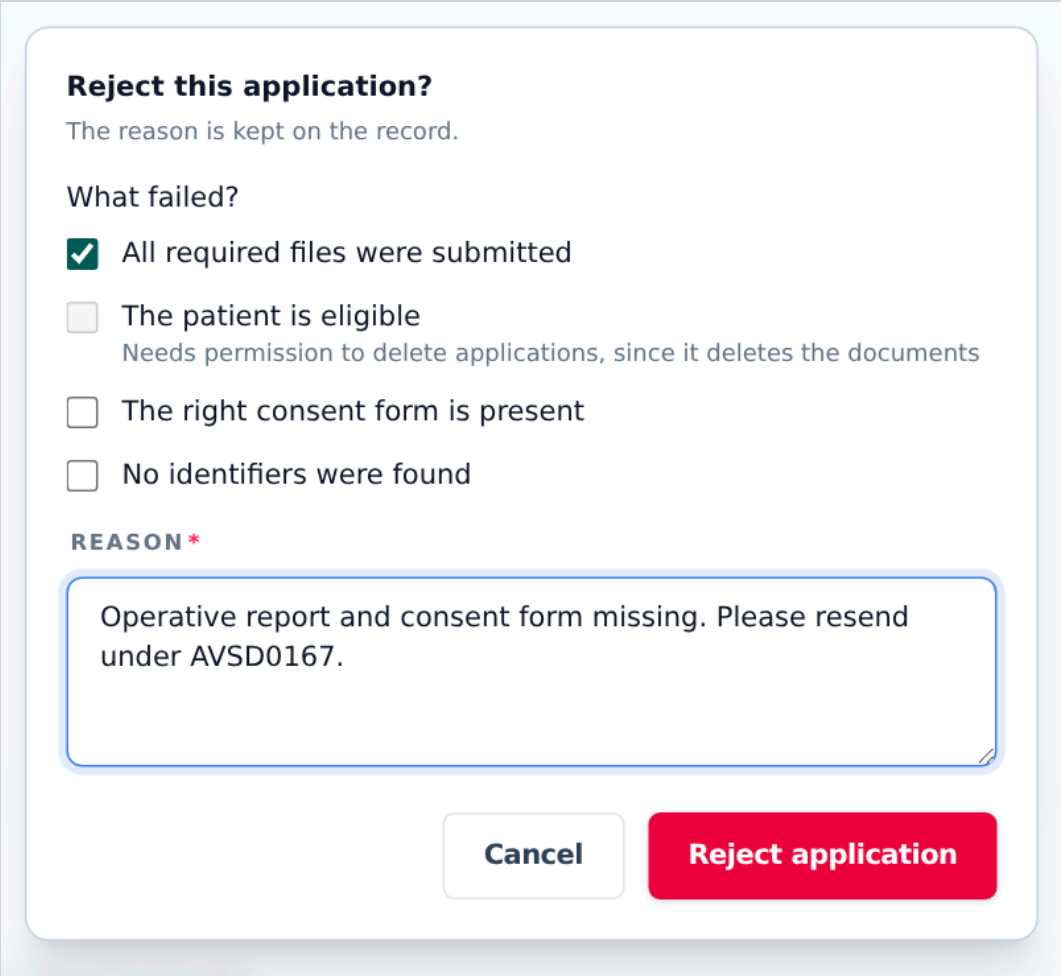


Figure 19. Rejecting an application: missing files

Failed point	What happens
1. Not all required files	Rejected. Files stay in Rejections so the missing ones can be added. Tell the site, quoting the patient code, so they resend.
2. Patient not eligible	All documents of the application (originals and copies) are **deleted at once** and the application becomes a deleted record. This cannot be undone. Only roles allowed to delete applications can choose it.
3. Consent form missing or wrong	Rejected. Tell the site, quoting the patient code, so they send the right form.
4. Identifiers found	Rejected. Redact again by hand (chapter 5.5) or ask the site to resend.

Protocol

§9.1.2: when an application is rejected the site coordinator is emailed with the reason; for reasons 1, 3 and 4 the email includes the UBRS so the site can resubmit. If the patient is not eligible (reason 2), all data is purged.

> **Not yet available in Hive**
> Hive emails the rejection (failed points and reason) to CRQ staff who get intake emails, not to the site. CRQ staff pass it on to the site coordinator.

5.4 Rejected documents

When one document is rejected (in step 2 or from the Zone 2 inbox), the rest of the application is kept. The application cannot be submitted until the document is fixed.

1. In step 2, click ⋯ → **Reject** on the document.
2. Write why (for example “Sonographer name still readable in footer”).
3. Click **Reject file**. The document now shows **rejected** and appears on the Rejections page.

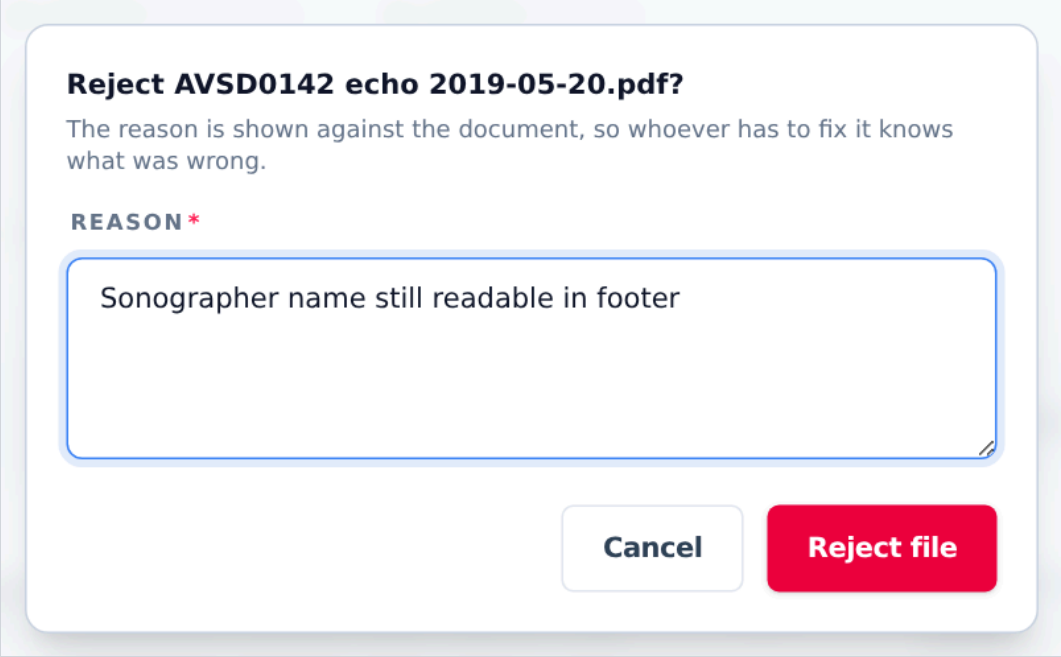


Figure 20. Rejecting one document

5.5 Rejections and manual redaction

The **Rejections** page lists everything turned down: **Rejected applications** at the top and **Rejected documents** below, each with the reason.

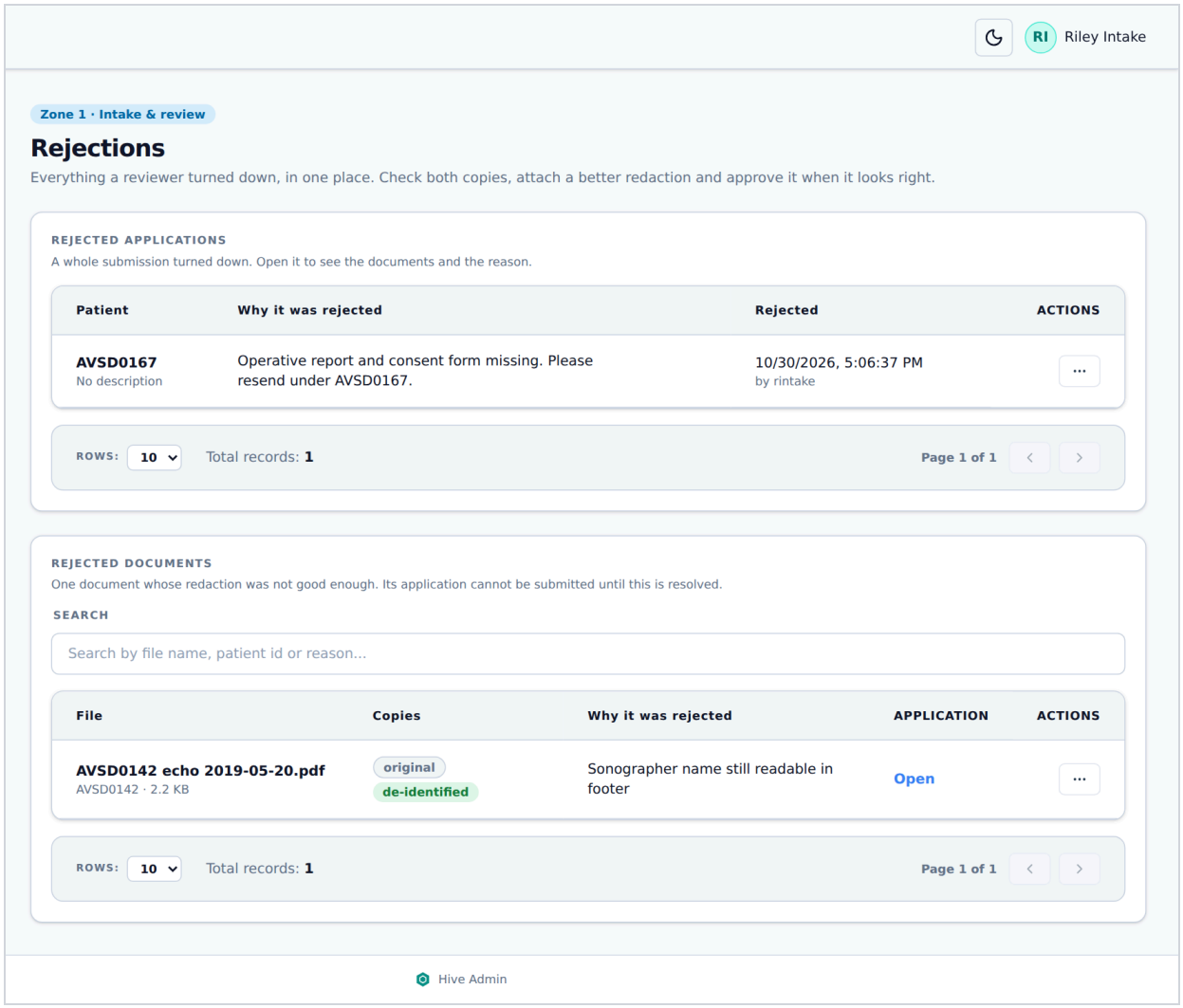


Figure 21. The Rejections page

Redacting a document by hand

1. On the rejected document, click ⋯ → **Download original** (or **Download de-identified**).
2. Open it in Adobe Acrobat Pro and redact what was missed with the Redact tool. Save it.
3. Back in Hive, click ⋯ → **Replace de-identified copy**, choose your file and click **Replace**.
4. The new copy keeps its place in the application, is named with **_RR** and is marked **by hand**. It goes back to review; check it and approve it.

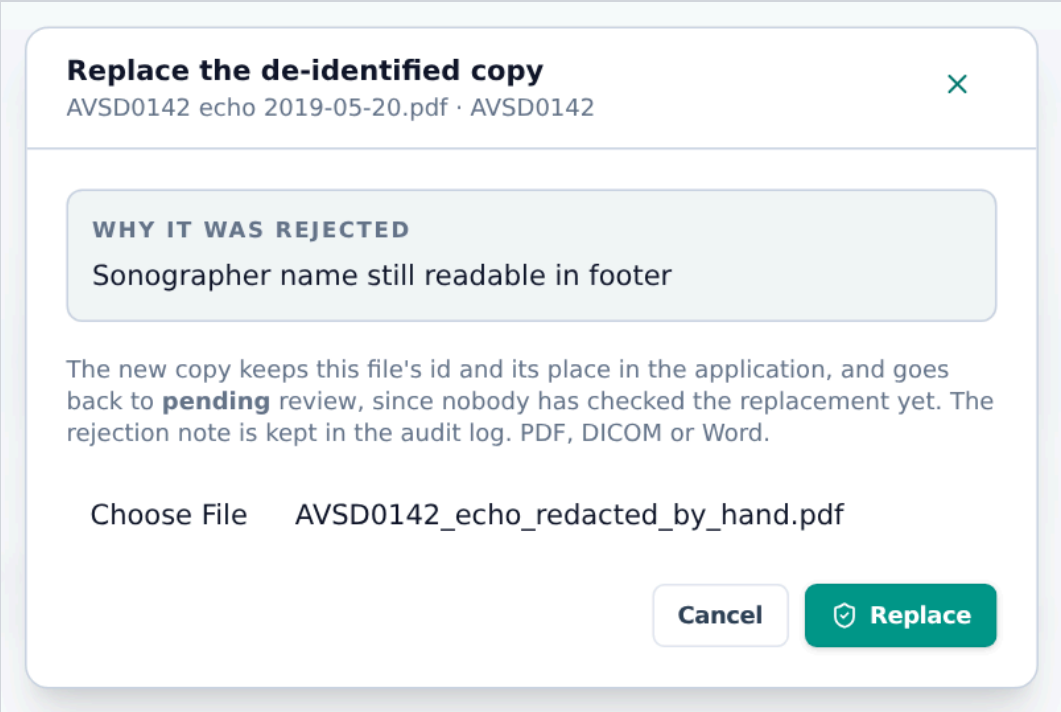


Figure 22. Replacing the de-identified copy with one redacted by hand

> **Protocol**
>
> §9.1.7: documents with remaining PHI are redacted manually with Adobe Pro, re-uploaded with an extra “R” in the suffix, and their redaction status is updated.

Other actions on the Rejections page: **View** and **Download** either copy, **Approve** (clear the rejection without replacing the copy), **Open application**, **Delete file** and **Delete application** (removes the files from disk for good; the record and reason stay).

5.6 Archived originals

After an application is submitted, its originals are kept in the archive and nobody can open them, not even staff who could before. The CHSS Manager can give one person access to one original, for example when the redacted copy looks damaged.

1. The CHSS Manager opens the submitted application, goes to **2. Documents**, and clicks ⋯ → **Access to the original…**.
2. In **Give access to**, pick the person. In **Why**, write the reason.
3. Click **Give access**. The person can now use **View original** on that document. Access can be removed in the same window.

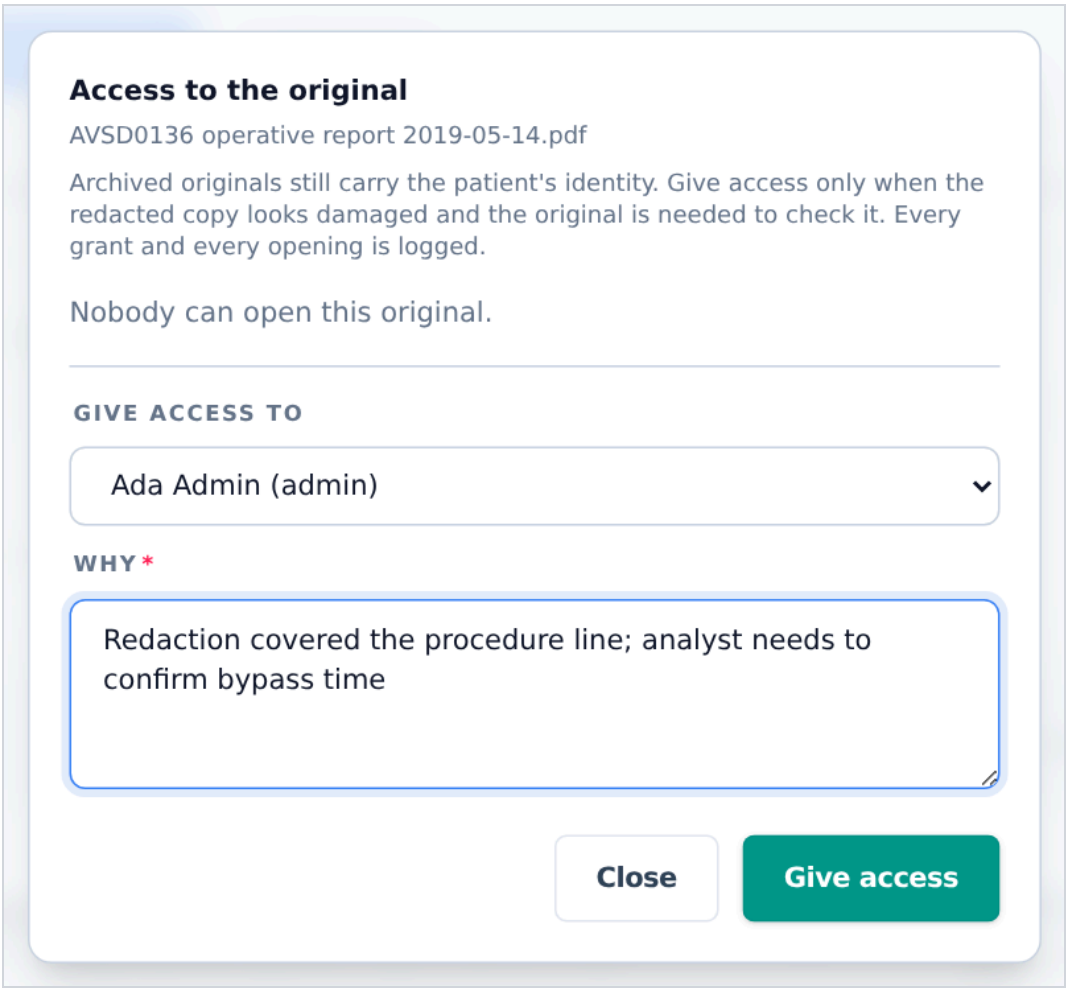


Figure 23. Giving one person access to an archived original

> **Protocol**
> §9.1.3: the archival storage holds all original PDFs. Access needs a specific permission granted only by the CHSS Manager, only for files suspected to be damaged by redaction. Every grant and every opening is logged (Audit log and Access log).

5.7 Files

The **Files** page lists every de-identified document across all patients. It is read-only; documents arrive through the De-Identifier. Use the search box to find a document by name, then **Open**, **Metadata** or (with permission) **Delete**.

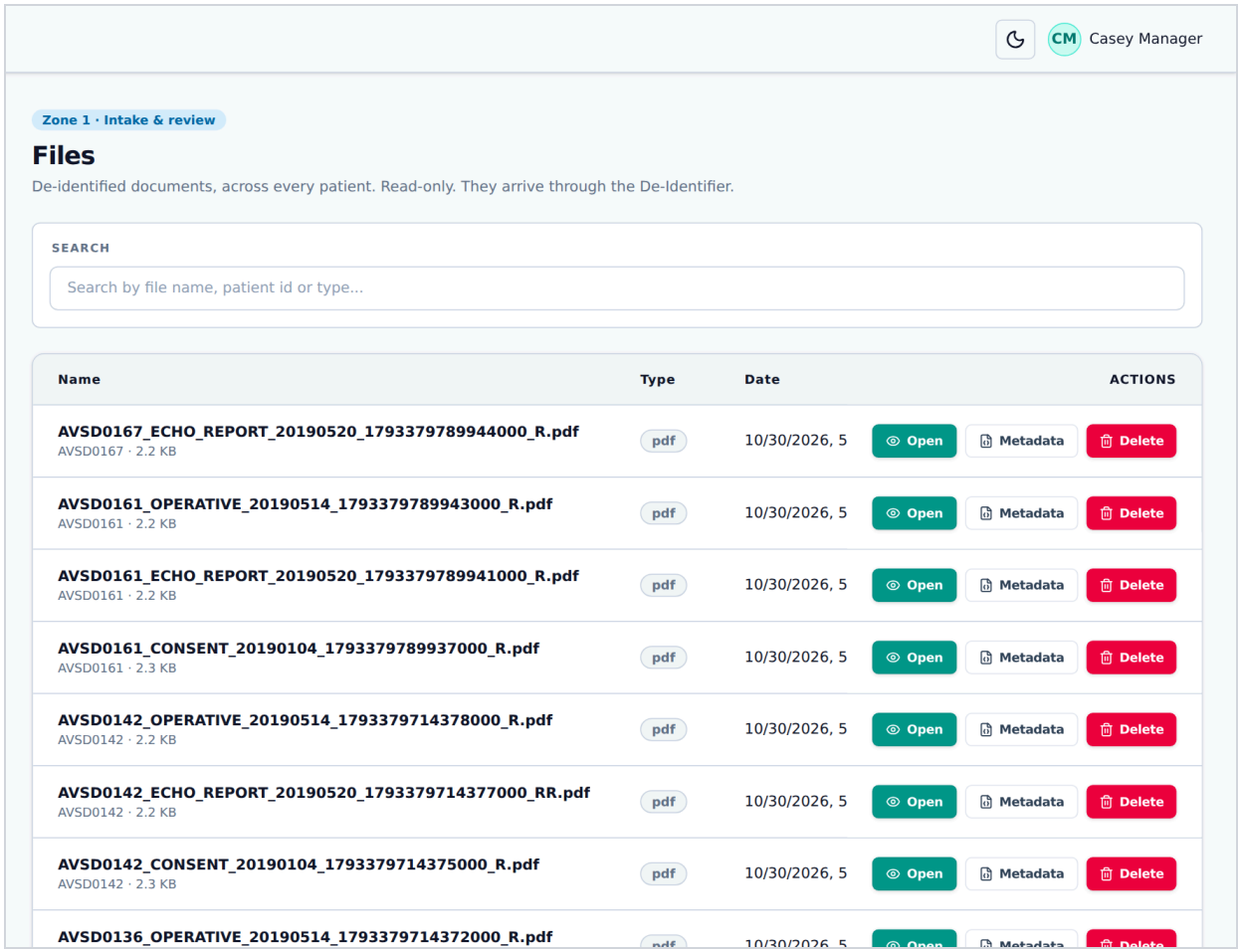


Figure 24. The Files page

> **Protocol**
>
> §9.1.5–9.1.6: the active storage holds every redacted PDF and is the main source for abstraction. Only senior analysts may see the whole of it.

5.8 Metadata

The **Metadata** page lists what Hive read from every document (type, dates, DICOM tags, PDF properties). Filter by **Status** and **Type**, or search. **Export to Excel** saves the list.

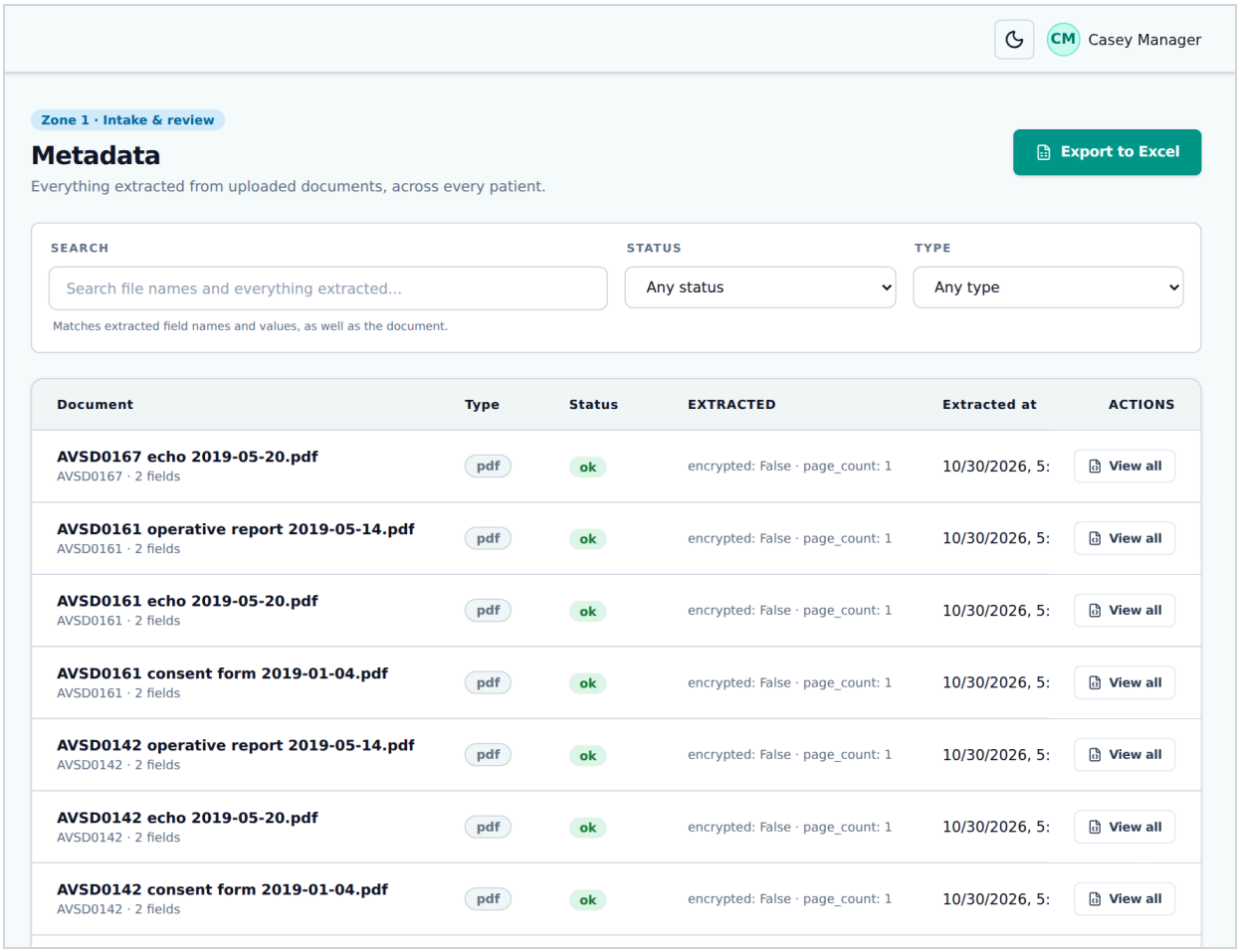


Figure 25. The Metadata page

> **Protocol**
>
> §9.1.2, §9.1.8: document and image metadata is stored so documents can be searched and retrieved.

5.9 Leaded images

Some images show identifiers in lead letters inside the picture, which cannot always be covered automatically.

1. In the application, click ⋯ → **Mark as leaded** on the image.
2. The image shows **leaded · not for Zone 2** in red. It can still be used in Zones 1 and 1X but can never go to Zone 2: preparing a safe copy with documents is refused while a patient in the cohort has one.
3. Once the image has been cleaned by hand and replaced, click ⋯ → **Clear leaded mark**.

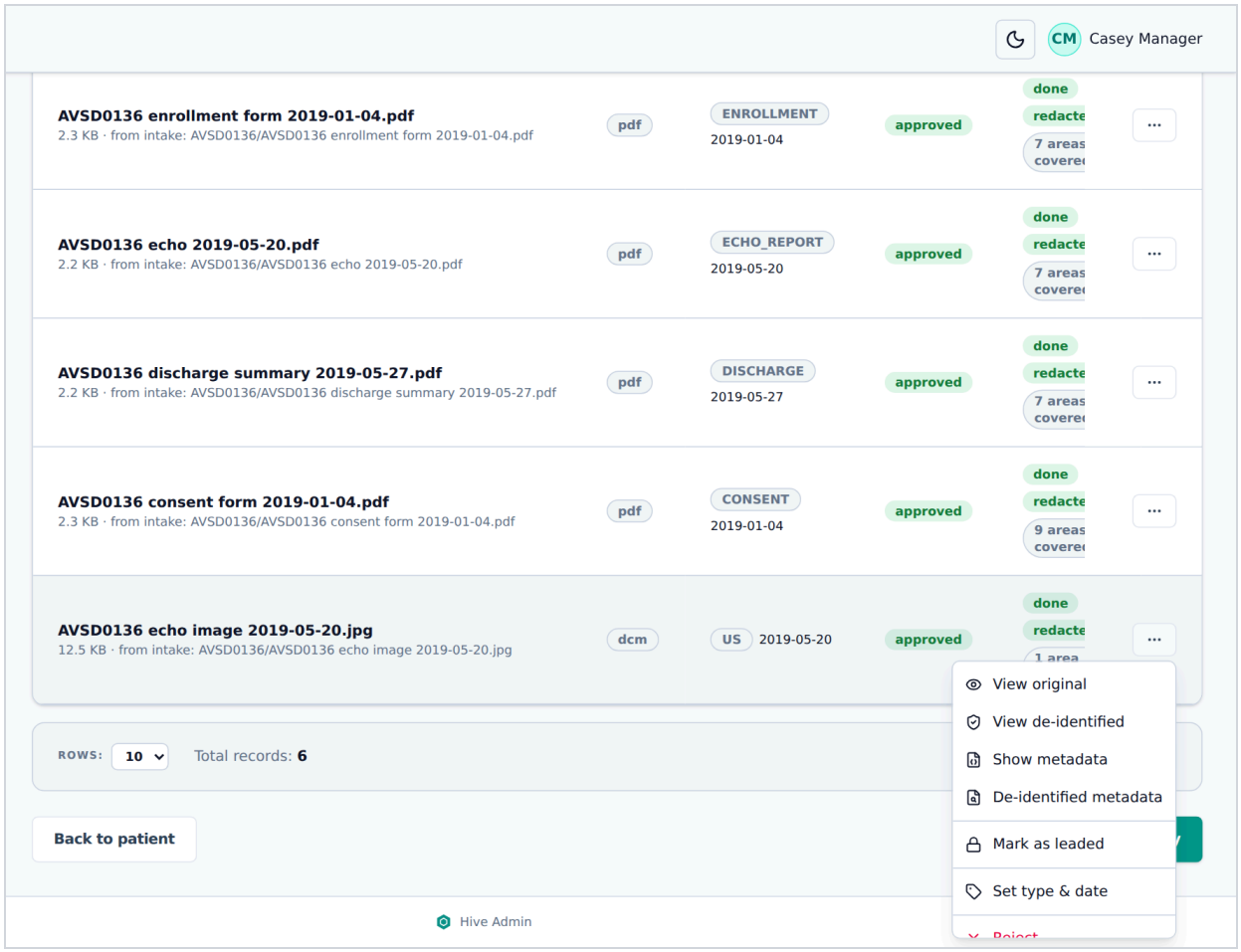


Figure 26. Mark as leaded, on an image

> **Protocol**
>
> §9.1.9: leaded images are marked in their metadata and kept in Zones 1 and 1X only.

> **Not yet available in Hive**
>
> §9.1.2: images and their metadata are sent to PACS. Hive keeps images in its own image storage; there is no PACS link.

5.10 Patients

The **Patients** page holds each patient's identity and contact details. This is the only place where names appear, and only in Zone 1.

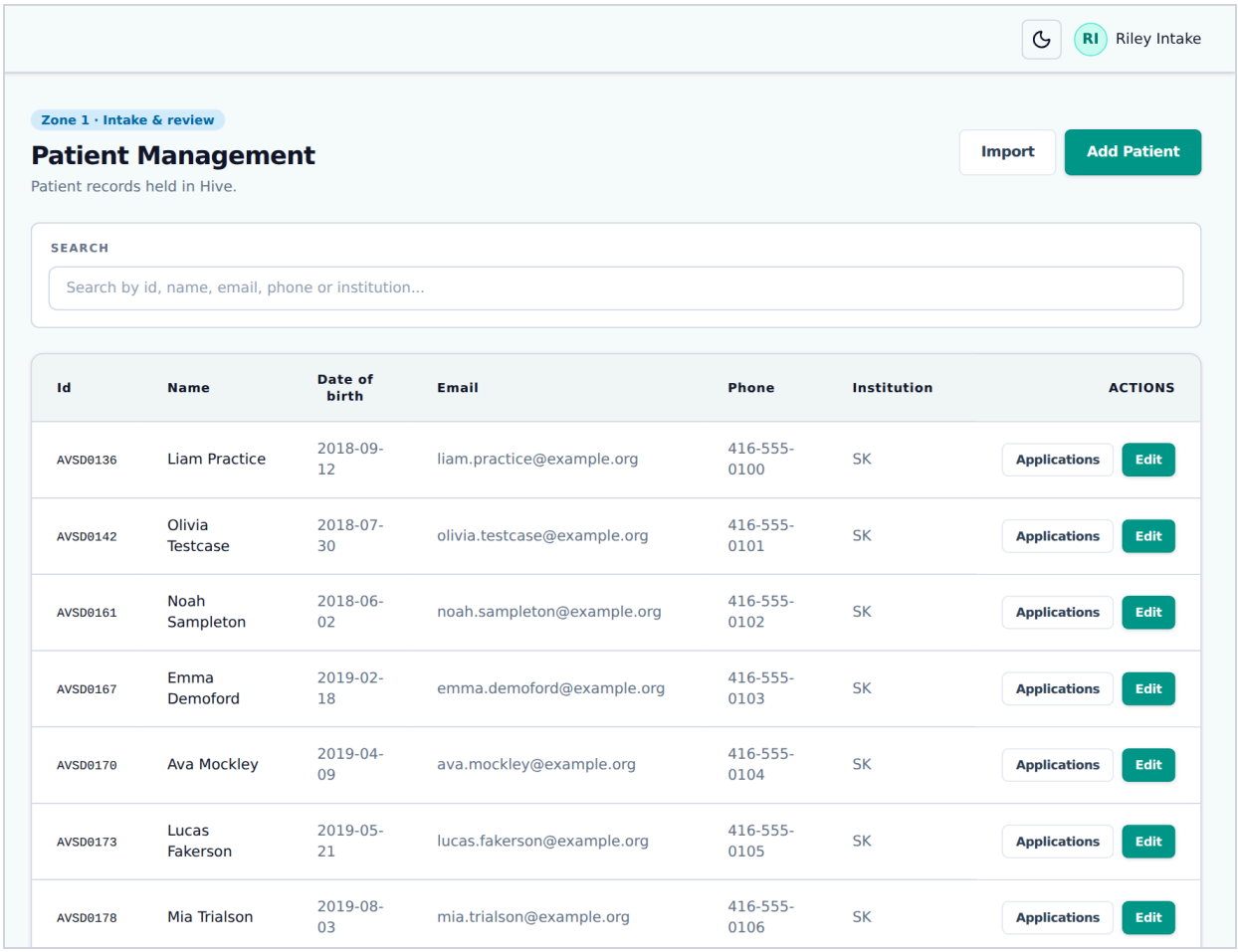


Figure 27. The Patients page

- **Add Patient** creates one patient. The code cannot be changed later.
- **Edit** changes the details. **Applications** lists the patient's applications.
- Tick **Site does not share records or contact details** for sites that only ask for a code (chapter 5.3).

Importing many patients

1. Click **Import**.
2. Click **Download template** to get the columns. Only **Patient code** is required; dates are Excel dates or YYYY-MM-DD.
3. Fill it in, choose the file (.xlsx or .csv) and click **Preview**. Nothing is changed yet. The preview shows what will happen to each row, and any problems or conflicts.
4. Click **Import N patients**. New codes are created; patients already in Hive only get their empty fields filled.

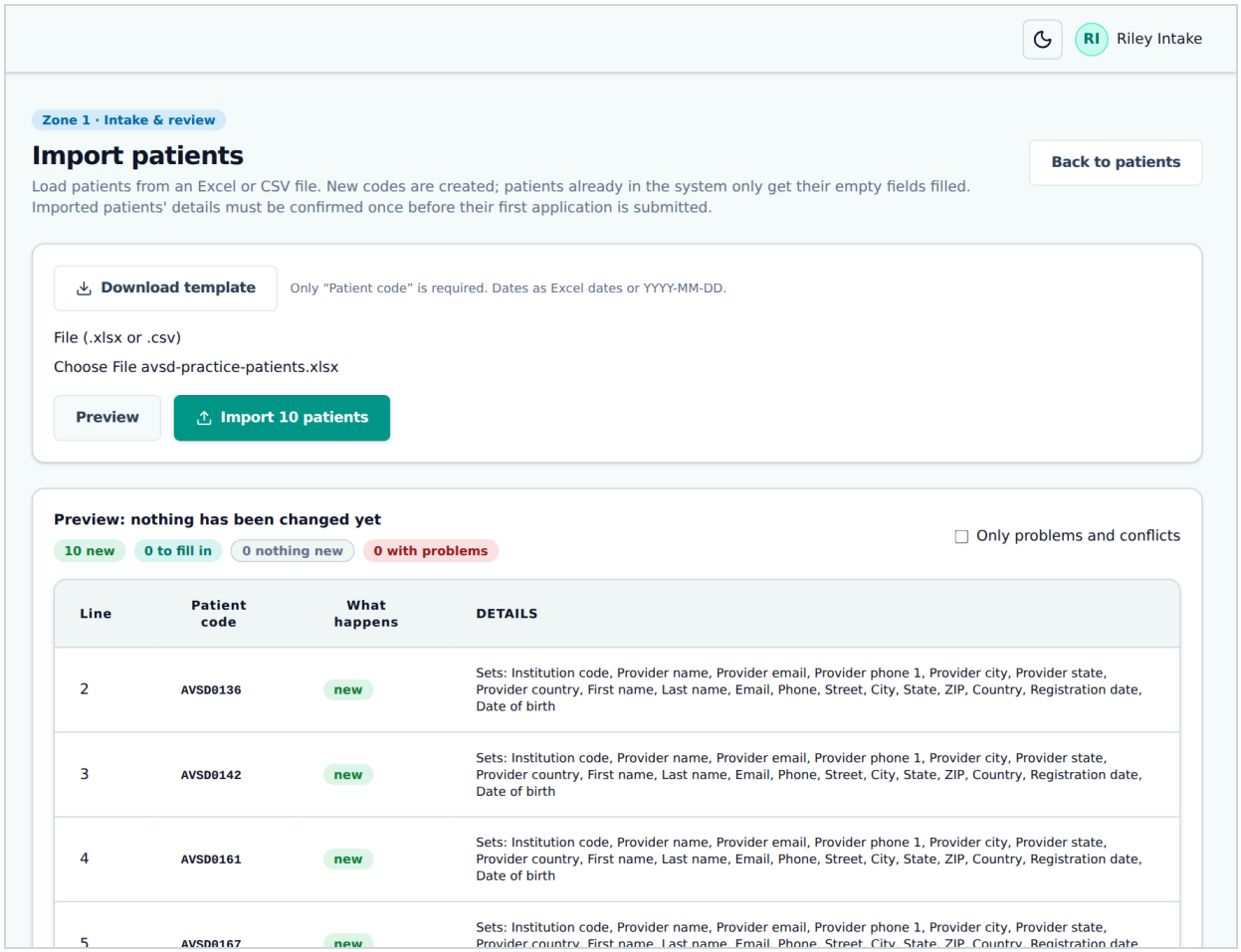


Figure 28. Preview of a patient import

> **Note**
>
> Imported patients must have their details confirmed once (**Details are correct**, chapter 5.3) before their first application can be submitted.

5.11 REDCap exchange

REDCap is the main tool for clinical data. The **REDCap** page brings REDCap data into Hive and sends lists back.

> **Protocol**
>
> §9.1.12: Python REDCap import APIs run every night and bring in the data, the data dictionary and the log (record creation and updates, queries, variable history).

> **Not yet available in Hive**
>
> The direct REDCap API connection is not set up yet. The page shows **Simulated: files**: someone with REDCap access exports the files from REDCap and imports them here. Once the API is connected, the same import runs every night by itself.

Importing from REDCap

1. In REDCap, export the **Data dictionary**, the **Records** (raw) and, if needed, the **Logging** as CSV.
2. In Hive, under **Import a REDCap export**, choose **What**: start with **Data dictionary**. It decides which fields may ever go to Zone 2 (coded, number and date fields; never identifier or free-text fields).
3. Enter the **Dataset (REDCap project)** name, for example avsd_clinical (one dataset per cohort, §9.5.2).
4. Choose the file and click **Preview**, then **Import**.
5. Repeat for **Records (data)**, then **Logging (audit trail)**. Records whose code is not a patient in Hive are skipped and listed.



Figure 29. Importing a REDCap export

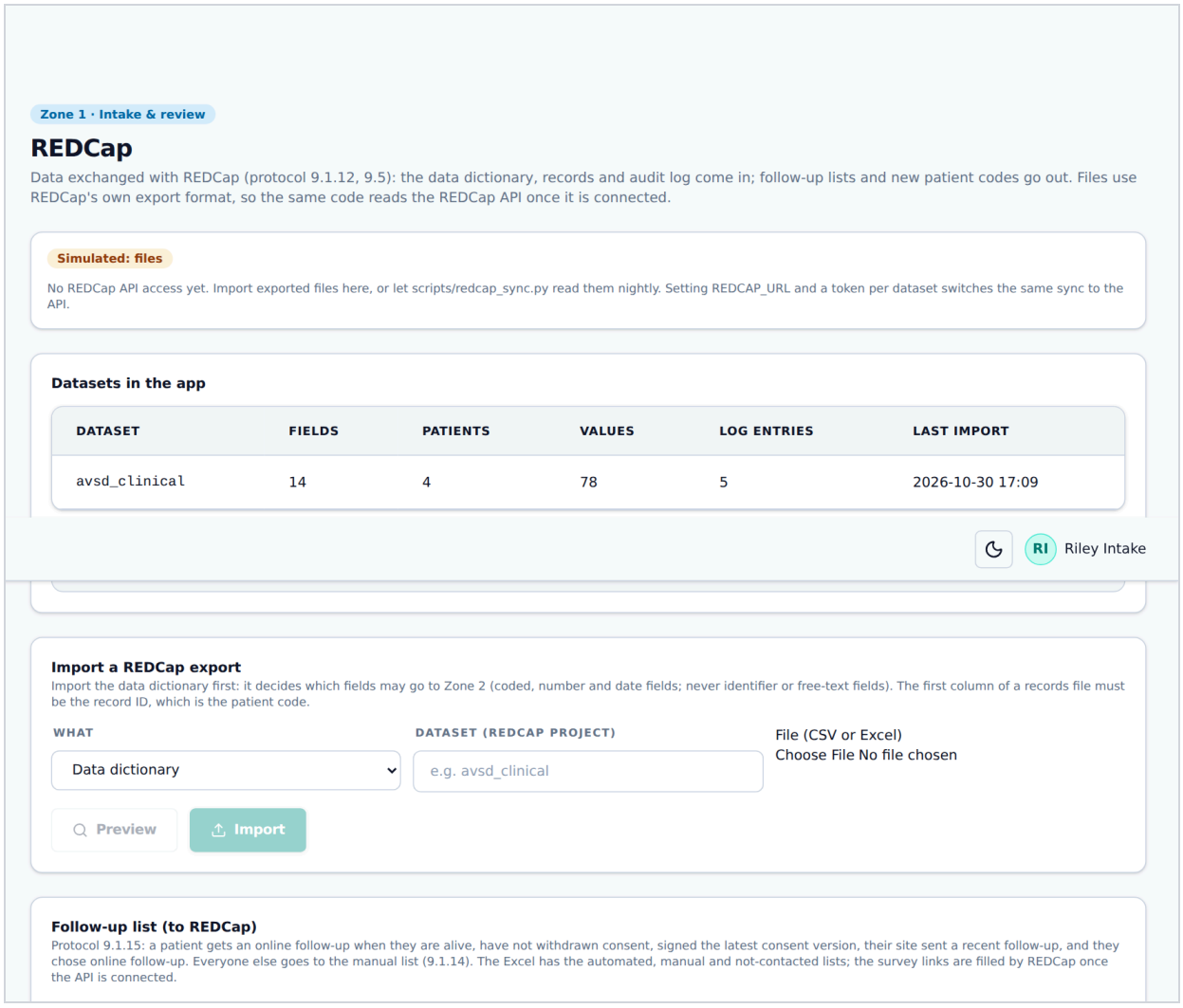


Figure 30. After the import: the dataset with its fields, patients and log entries

New patient codes (to REDCap)

Lists patients in Hive with no record yet in a REDCap dataset, as a REDCap import file. Enter the **Dataset**, click **Download CSV** and import it in REDCap, so sites enter data only against a valid code.

> **Protocol**
>
> §9.1.2: no data is entered without a valid UBRS.

5.12 Follow-up

Patients are followed up periodically, online where possible.

Protocol

§9.1.15: a patient gets an automated online follow-up when they are alive and have not withdrawn consent, and (1) signed the latest consent version, (2) their site's follow-up is current, and (3) they chose online follow-up in the consent matrix. They receive a personalised REDCap link by email.

§9.1.14: everyone else goes on the manual follow-up list.

1. On the **REDCap** page, in **Follow-up list (to REDCap)**, enter the **Latest consent version** (for example v3) and how many days count as a **Recent follow-up from the site**.
2. Click **Check** to see how many patients are automated, manual or not contacted, and why.
3. Click **Download list (Excel)**. It has the **Automated**, **Manual** and **Not contacted** lists, with the reason for each patient and whether a consent update is due (for example turning 18).

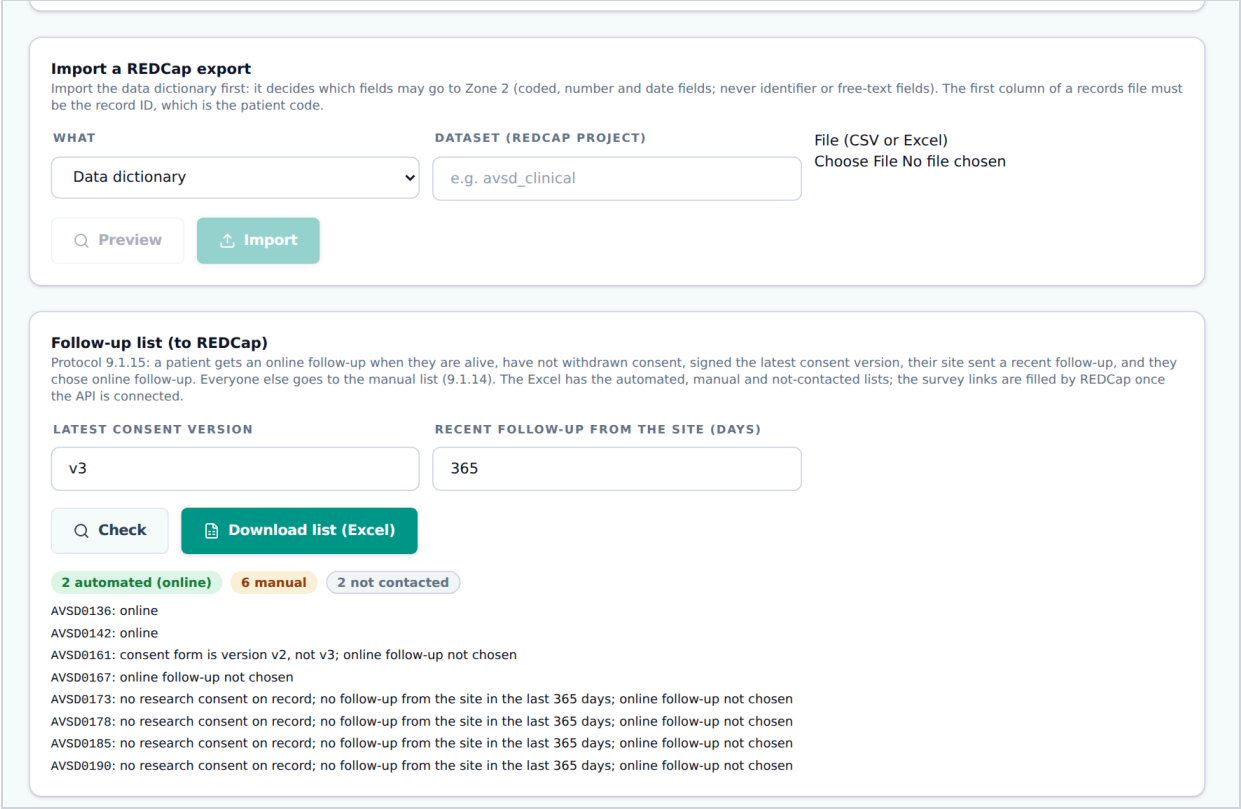


Figure 31. Checking who gets an online follow-up

Not yet available in Hive

Sending the follow-up: the personalised REDCap survey links and invitation emails (§9.5.4), and running the check on a schedule.

Manual follow-up (§9.1.14): pulling contact details and printing personalised forms into a follow-up folder.

Today Hive produces the lists only.

[Picture to be added]

Picture of the REDCap invitation email and survey, once the REDCap API is connected.

Figure 32. REDCap follow-up survey invitation

5.13 Data quality check

> **Protocol**
>
> §9.1.13: after each import an automatic Python quality check looks for missing critical variables, impossible values and conflicting information. It produces a PDF report per cohort for the designated staff. Issues that were checked and accepted are suppressed so they are not raised again.

Hive's data quality check runs as a scheduled job (not a page). It checks patients (for example missing date of birth, death before birth, age over 100), applications (submitted without documents or checklist), documents (unknown type or date, dated before birth or after death, failed de-identification) and consent (no research consent, withdrawn consent in an active project). The report comes as PDF and Excel, and staff are emailed the counts. Accepted issues are listed in a suppression file so they are not raised again.

> **Not yet available in Hive**
>
> A page in Hive to see the report, one report per cohort, and checks that compare Hive with REDCap.

[Picture to be added]

Picture of the first page of the data quality PDF report.

Figure 33. Data quality report

6. Zone 1X: preparation

Zone 1X is where data is prepared for one approved project. It holds de-identified data with CHSS codes, so nothing in Zone 1X may leave it. Only the safe copy (chapter 6.3) goes on to Zone 2.

6.1 Consent matrix

The **Consent matrix** records each patient's consent decisions. Hive applies it everywhere: a patient whose research consent is refused or withdrawn cannot be in any project; for projects approved for PDFs and images, the same applies to the documents consent.

Consent	Meaning
Research use of data	The patient's data may be used for research
PDFs and images	The patient's documents and images may be used
Online follow-up preferred	The patient chose online follow-up (chapter 5.12)

Each decision is **granted**, **refused** or **withdrawn**, with the form version and date. Decisions are never changed or deleted; the latest one of each type counts and earlier ones are kept as history (**Show the full history**).

Recording one decision

1. Under **Record a decision**, enter the **Patient code**, pick the **Consent** and the **Decision**.
2. Enter the **Form version**, the **Consent date** and, if useful, a **Note**.
3. Click **Record**.

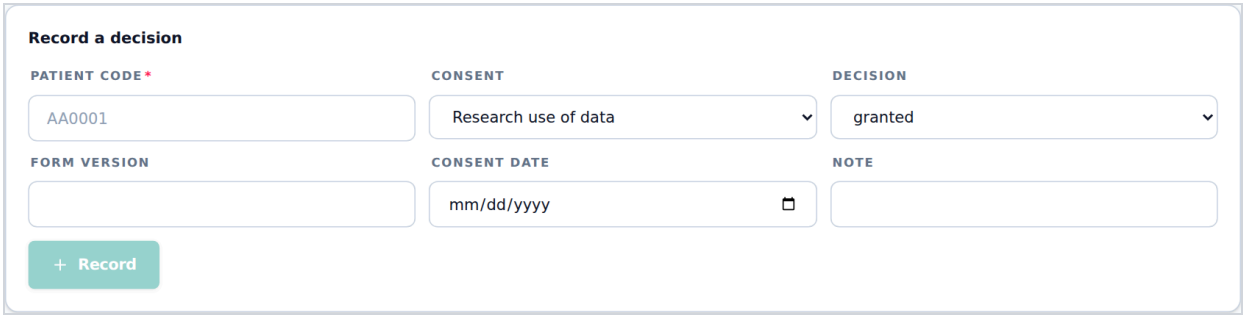


Figure 34. Recording a consent decision

Importing from a spreadsheet

1. Under **Import from a spreadsheet**, click **Template** for the columns: Patient code, Consent type, Status, Version, Consent date, Note.
2. Choose the file and click **Preview**. Rows that match the latest decision are skipped.
3. Click **Record N decisions**.

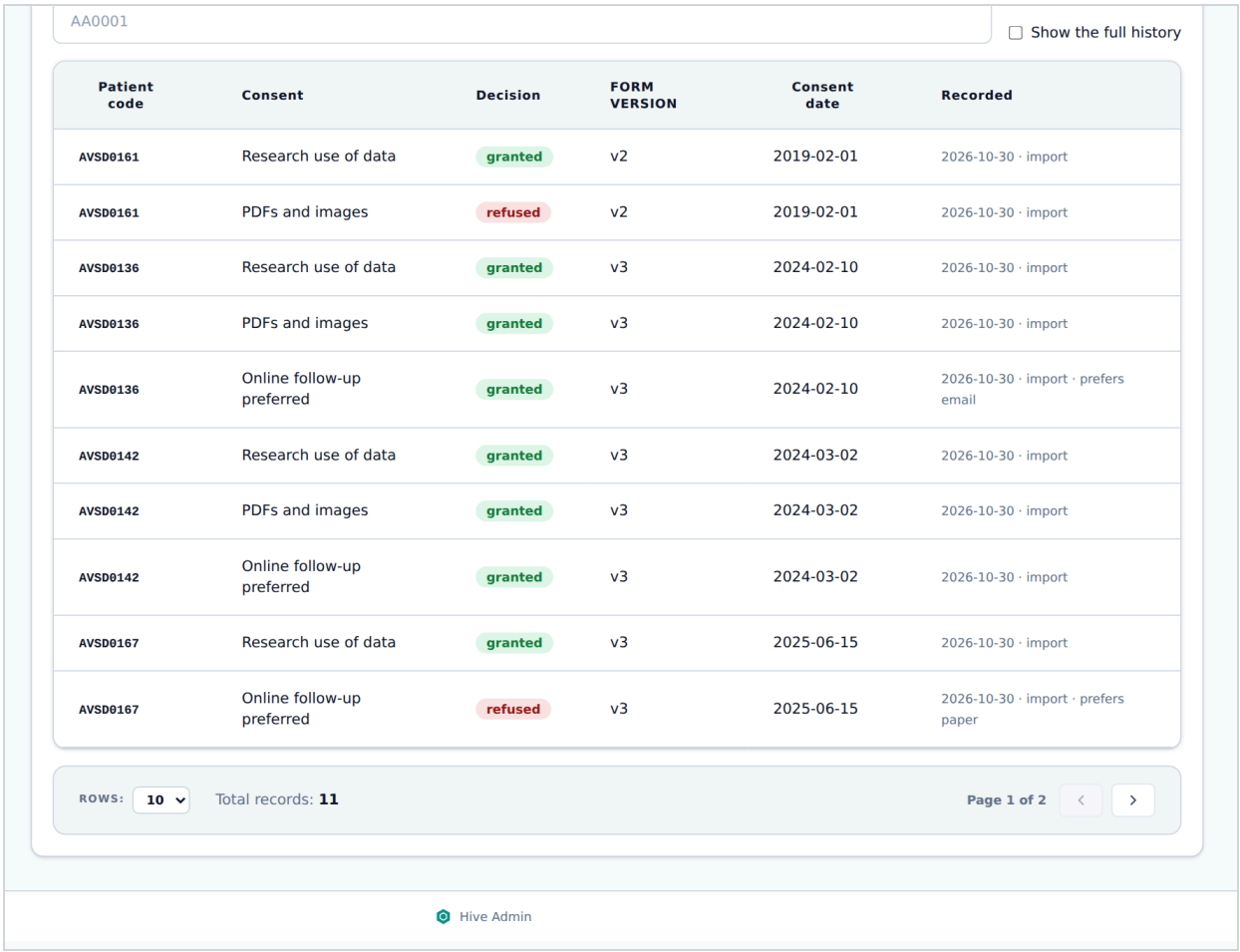


Figure 35. The consent matrix

Protocol

§9.5.1: the consent and contact matrix is kept in REDCap in a machine-readable form, with each patient's consent status, version and restrictions, and is applied whenever data is extracted.

Not yet available in Hive

Reading the consent matrix straight from REDCap (today it is imported from a REDCap export).

§6.7: when consent is withdrawn, the patient's documents and copies are deleted. Today Hive keeps a withdrawn patient out of every new project, workspace and release, but does not delete their documents.

6.2 Abstraction folders

An abstraction folder is a temporary, read-only copy of de-identified documents for one person's task, for example abstracting echo results into REDCap. Only documents of submitted applications can be used.

1. Open **Abstraction folders** and click **New folder**.
2. Describe the documents you need. Leave a field empty to take any value: **Patient codes**, **Types / modalities** (for example ECHO_REPORT, MR), **Document date from** / **to**, **Text in the file name or metadata**, and the file types.
3. Click **Find documents**. Hive shows how many documents for how many patients match.
4. Enter a **Folder name** and the **Purpose**. In **Assign to**, pick the person doing the work, or leave **Myself**.
5. Click **Make folder**. The folder location is shown in the list; only you and the assigned person can open it.

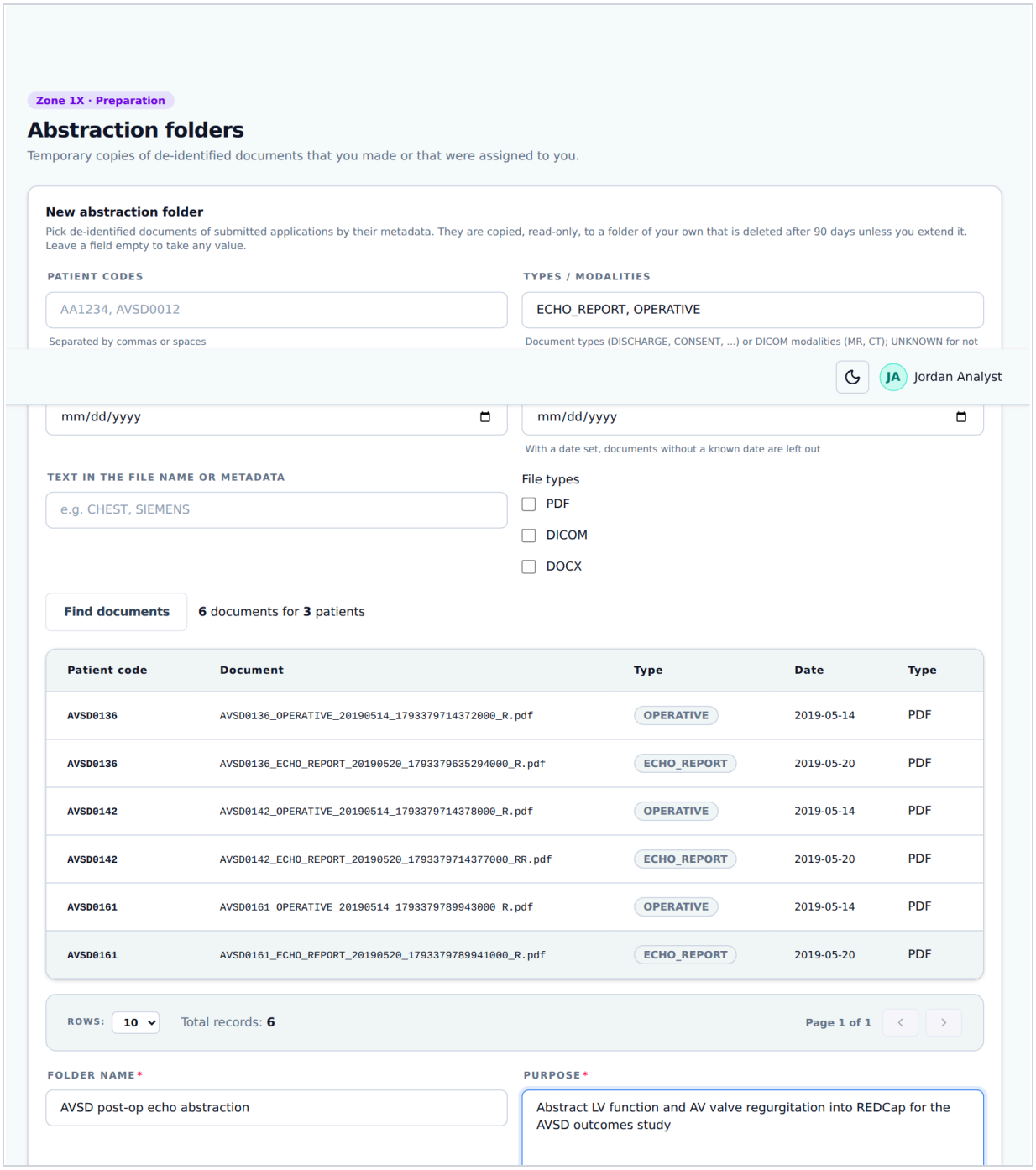


Figure 36. Making an abstraction folder

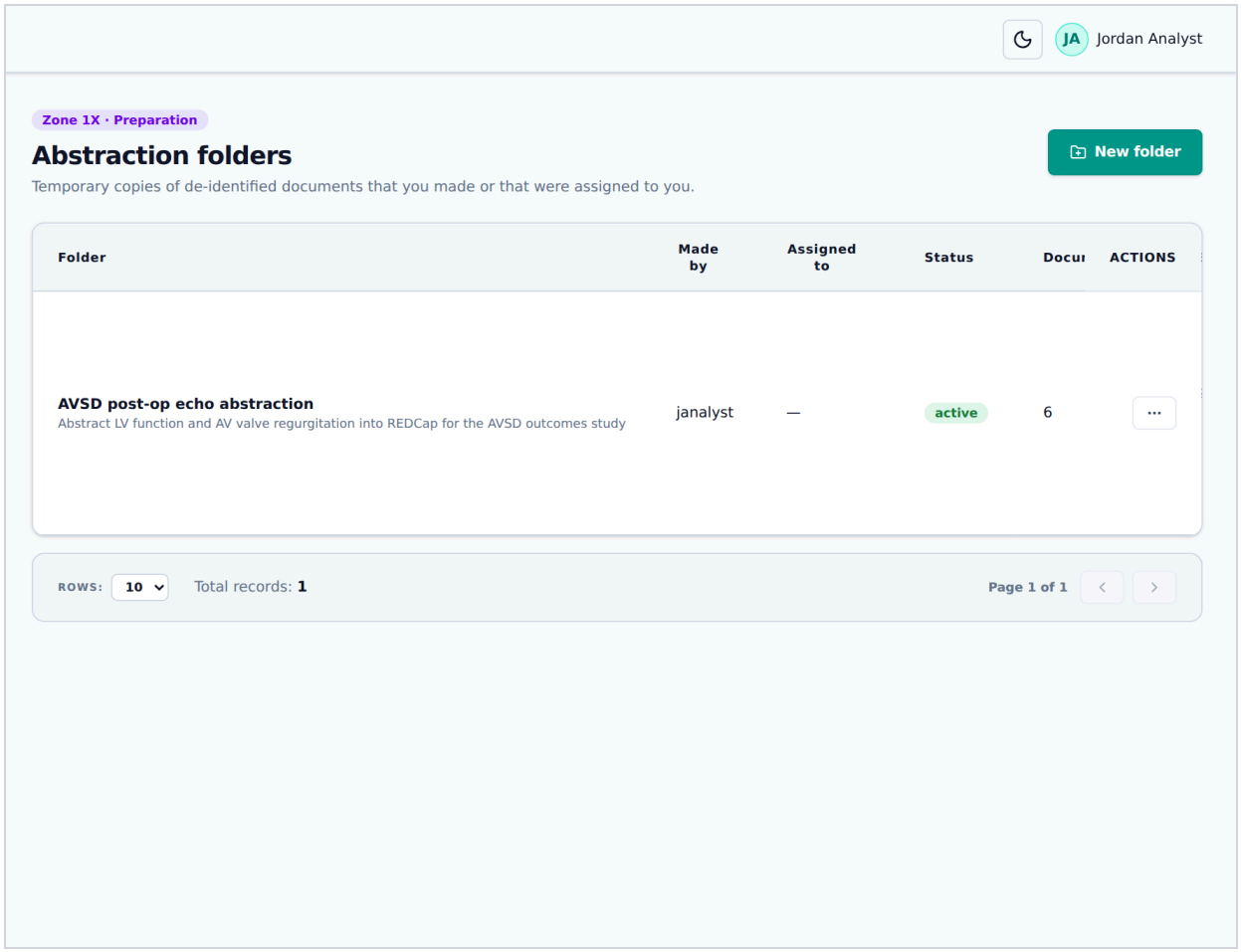


Figure 37. An active abstraction folder, deleted automatically after 90 days

Click ⋯ on a folder for: **Show documents**, **Extend 90 days**, **Copy again** (copies the same documents with a new 90 days) and **Delete** (removes the copy; the record stays).

> **Protocol**
>
> §9.1.6, §9.1.9: documents needed for a task are copied to a temporary folder chosen by their metadata, accessible only to the end user. No new data is added to it. It is deleted when the task is done, or automatically after 3 months unless extended, and can be regenerated.

6.3 Projects

A project is an approved study that needs data. Each project picks its patients, has its own workspace in Zone 1X, and ends with a safe copy released to Zone 2. Only members can open a project.

[Picture to be added]

Copy Figure 6 of the protocol here: request received → check approvals → define cohort → check consent → extract → clean → Zone 2 preparation → manager review → release.

Figure 38. Zone 2 data release workflow (protocol Figure 6)

Creating a project

1. Open **Projects**. Under **New project**, type the name and click **Create project**. Hive gives it a code (for example P2BE6) and opens it.
2. In **Project and approvals**, enter the **Approval reference** (the ethics or data access approval, for example the REB number) and a **Description**.
3. Tick **Patient consent confirmed** once consent has been checked. Patients refused or withdrawn in the consent matrix are always kept out.
4. Tick **Approved for documents (PDFs and images)** only if the project has a demonstrated need for the files (for example machine learning). Without it, the safe copy lists the documents but does not include them.
5. Click **Save**.

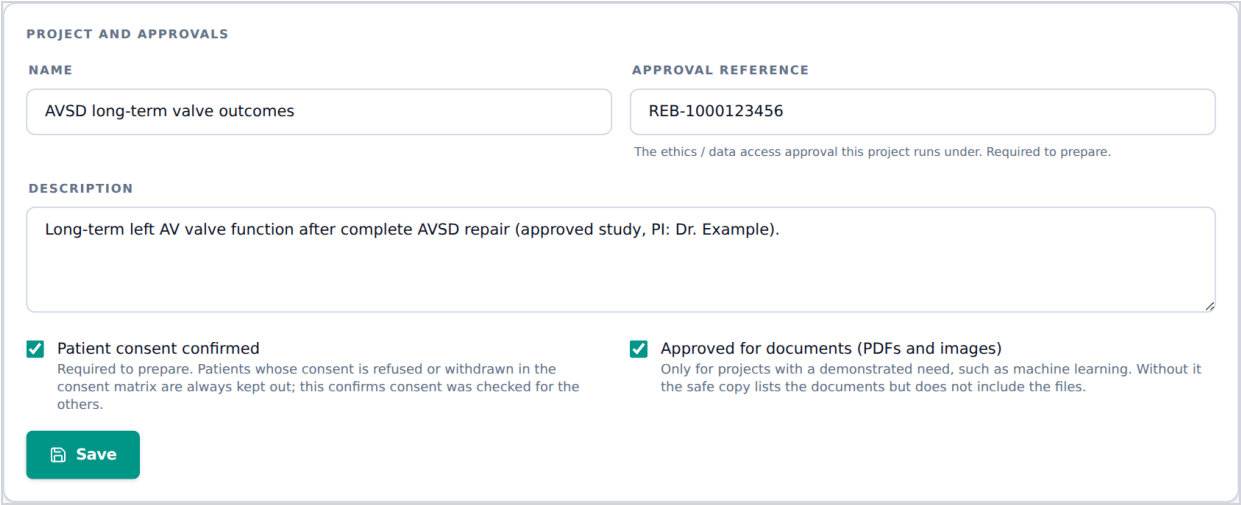


Figure 39. Project and approvals

Protocol

§9.3 (Figure 6): a request is checked for its approvals; an incomplete request is returned to the requester. The approvals (REB, agreements, training) are checked by the CHSS Manager outside Hive.

Members

Under **Members**, pick a person in **Add a member** and click **Add**. Only members can open the project and its workspace. Click **Remove** to take someone off. Adding and removing members needs the right to manage all projects (CHSS Manager or Application Specialist).

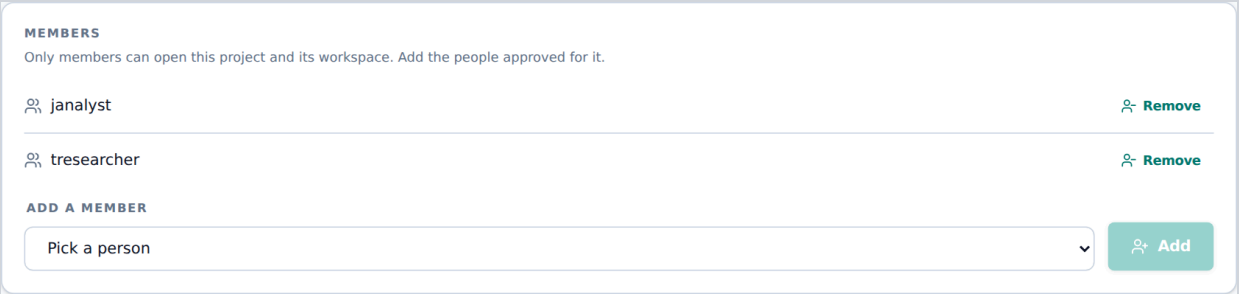


Figure 40. Project members

Choosing the patients (cohort)

1. Under **Patients**, click **Choose patients**. Only patients with at least one submitted application are offered.
2. Tick patients by hand, or click **Find by criteria** to select them by inclusion and exclusion criteria: **Cohort (code prefixes)**, **Registered from / to**, **Age at registration**, **Institution codes**, **Countries**, **States / provinces**, **Has documents of type or modality**, **Document dated from / to**, **Exclude these codes** and **Living patients only**.
3. Click **Find**. Hive shows the matches and, separately, patients that matched but are kept out by the consent matrix, with the reason.
4. Click **Select only the matches** (or **Add matches to the selection**), then **Save patients**.

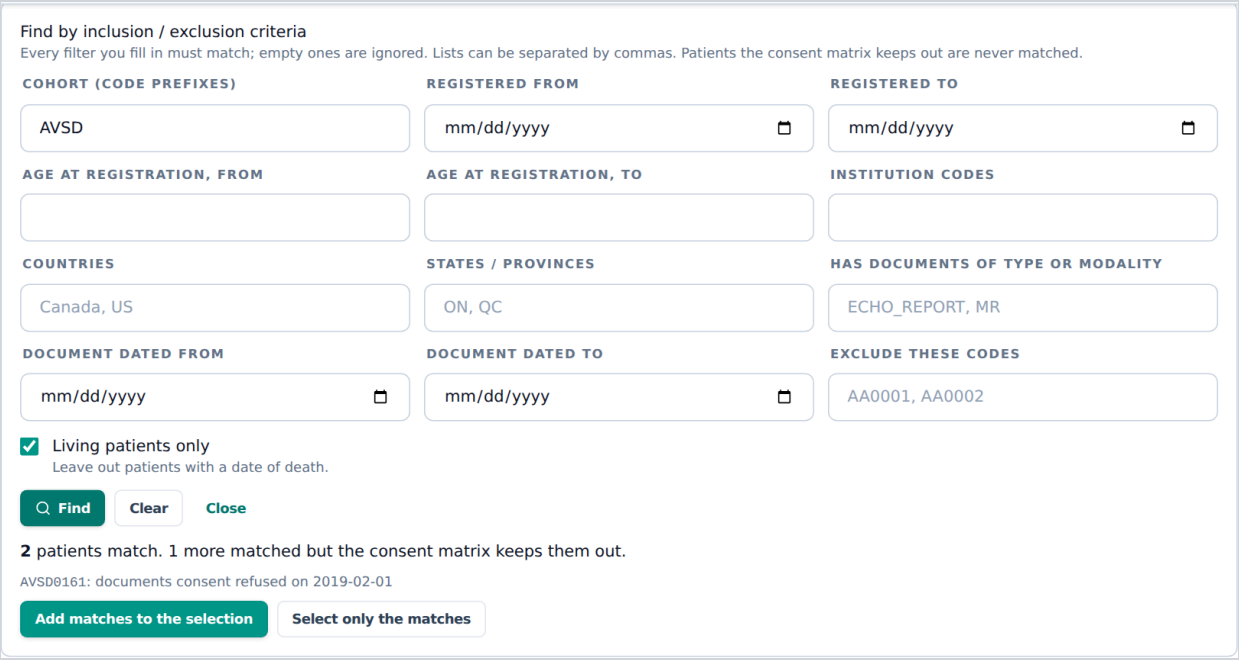


Figure 41. Finding patients by criteria; one patient is kept out by the consent matrix

Each patient gets a project-only ID (for example P2BE6-FE77G5) and a random date shift. Both are kept in Hive and never released.

> **Protocol**
>
> §9.2.1: the cohort explorer selects patients by named cohort or by inclusion and exclusion criteria, and the consent matrix removes patients who did not authorise the use.

Workspace (Zone 1X)

The workspace is the project's own folder for data cleanup. It holds the cohort's tables and, if wanted, their de-identified PDFs and images. CHSS codes stay in it, so nothing in the workspace goes to Zone 2.

1. Tick **Copy the PDFs and images** if the project needs the files.
2. Click **Build workspace**. The folder location and date are shown. **Rebuild workspace** builds it again after changes.

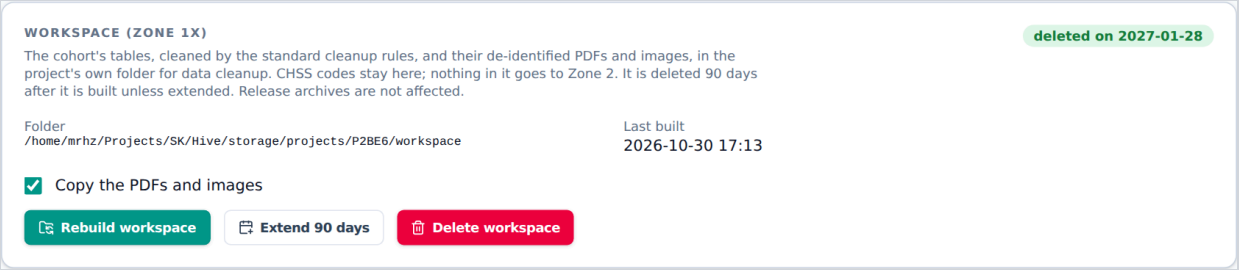


Figure 42. A built workspace

File	Contents
patients.csv, applications.csv, documents.csv	The cohort, their submitted applications and documents (no names or contact details)
redcap_<dataset>.csv	REDCap data of the cohort, one file per dataset (identifier fields left out)
documents/	The de-identified files, by patient code (if copied)
cleaning_log.csv	The standard cleanup rules applied and how many rows each changed
profile.xlsx	For each column: missing values, distinct values, range, frequencies and outliers

The workspace is deleted 90 days after it was first built. To keep it longer, click **Extend 90 days** and record which approval allows it. **Delete workspace** removes it now; release archives are not affected.

> **Protocol**
>
> §9.2.2: project storage holds the patients' data and, if needed, their PDFs and images. It is provided by the Application Specialist or Manager and kept 90 days unless extended by approval.
>
> §9.2.3: correct data in the right place, never in Excel. (1) A single error is fixed at the source (REDCap). (2) A recurring error is fixed by a rule in the dataset builder, applied at every build. (3) An analysis-specific change goes in the analysis-ready script.

> **Note**
>
> Recurring corrections (rule 2) are written as cleaning rules by the Application Specialist and kept in GitLab. Ask them when the same error keeps appearing.

Safe copy (Zone 2)

Prepare safe copy makes the data that may go to Zone 2. It needs an approval reference, consent confirmed and at least one patient.

- Remaining PHI is removed: names, addresses, phone numbers, emails and dates of birth are left out (age at registration is given instead).
- Every date of a patient is shifted by that patient's own random number of days, between −1826 and +1826 (up to 5 years either way). Intervals between dates stay true.
- Free-text fields are removed; only coded, number and date fields defined in the data dictionary go.
- PDFs and images are included only if the project is approved for documents, and never leaded images.
- Every patient, application and document gets a project-specific ID that cannot be matched to the CHSS code.
- A **data_dictionary.csv** explains every column.

1. Click **Prepare safe copy**.
2. If a patient is kept out by the consent matrix, Hive refuses and names the patient. Take them out of the cohort and try again.
3. The release appears with status **waiting for review**. It is now in the Zone 2 inbox for the CHSS Manager.

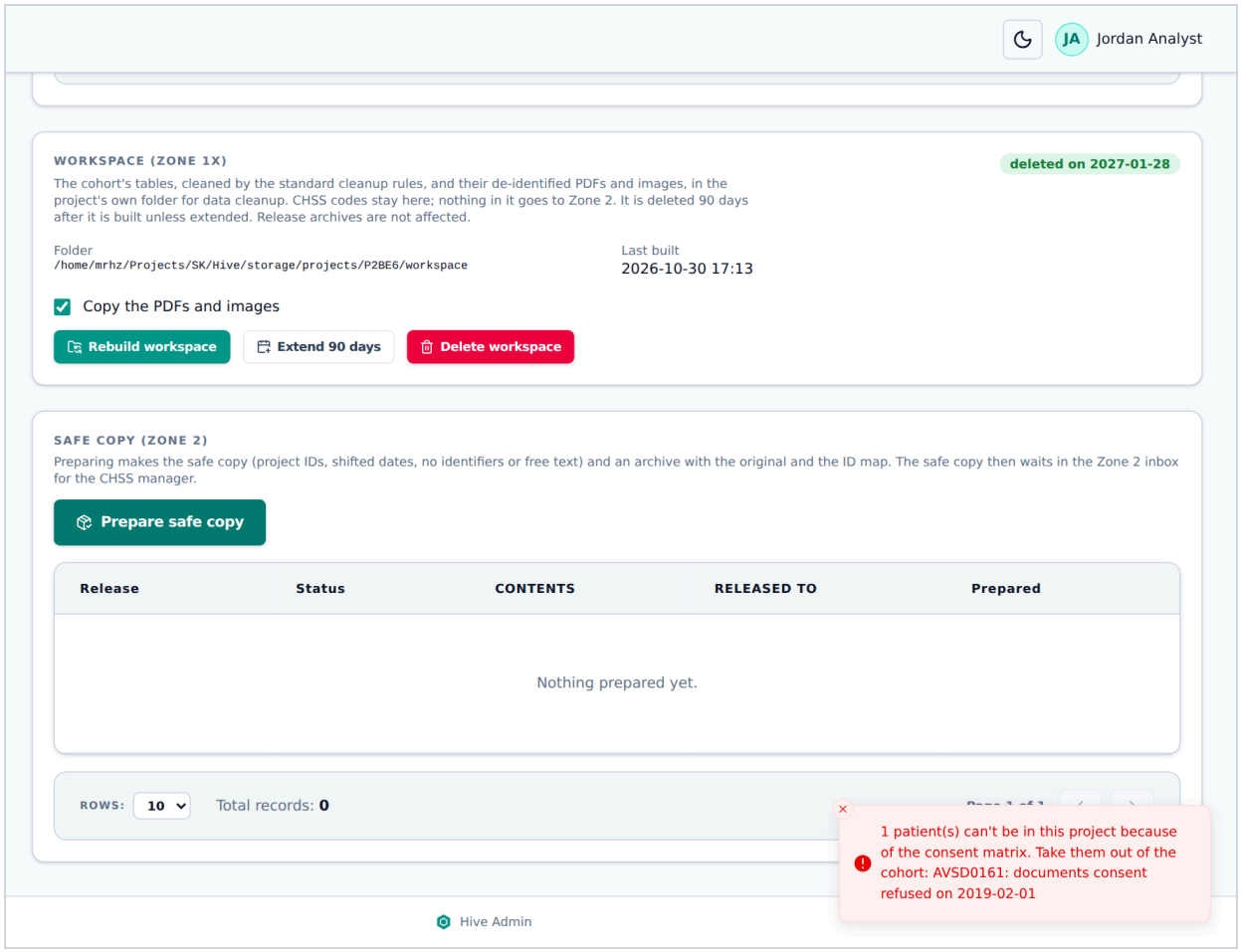


Figure 43. Prepare is refused: one patient refused the documents consent

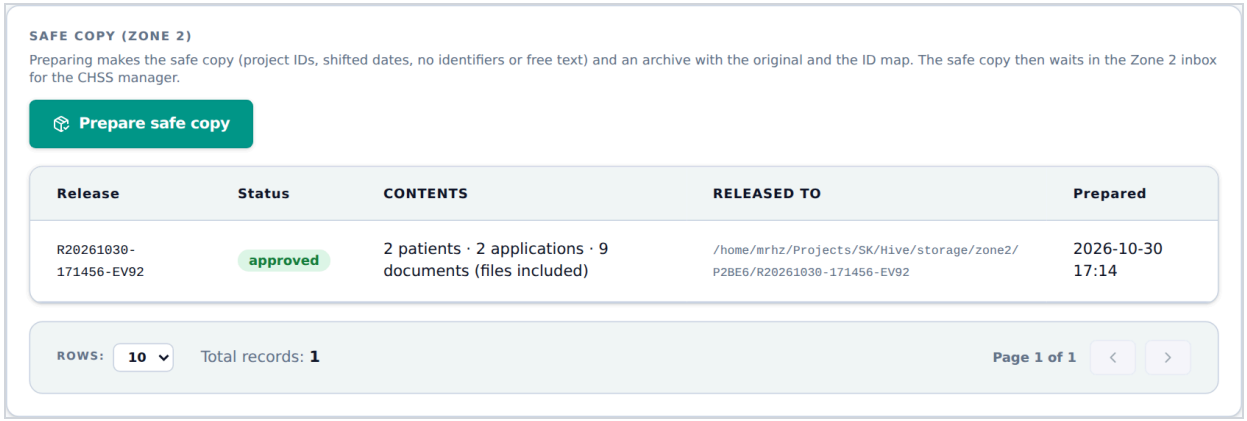


Figure 44. The safe copy after release

Protocol

§9.2.4: the Zone 2 preparation script is applied to every release: (1) remove remaining PHI, (2) shift dates by a random whole number of days per patient between −1826 and +1826, (3) remove free-text fields, (4) delete PDFs and images unless approved, (5) create project-specific patient IDs. The original data, the ID map and the approved data are archived.

Ending a project

When the study is finished, click **End project** and give the reason. No more safe copies can be prepared. The Zone 2 copy and the archive stay. Remove the researchers' access in **Members** and on the **Users** page.

7. Zone 2: release and analysis

7.1 Zone 2 inbox (CHSS Manager)

The **Zone 2 inbox** holds safe copies waiting for the CHSS Manager. Check that they are free from identifiers and that the preparation was applied properly, then release them or send them back.

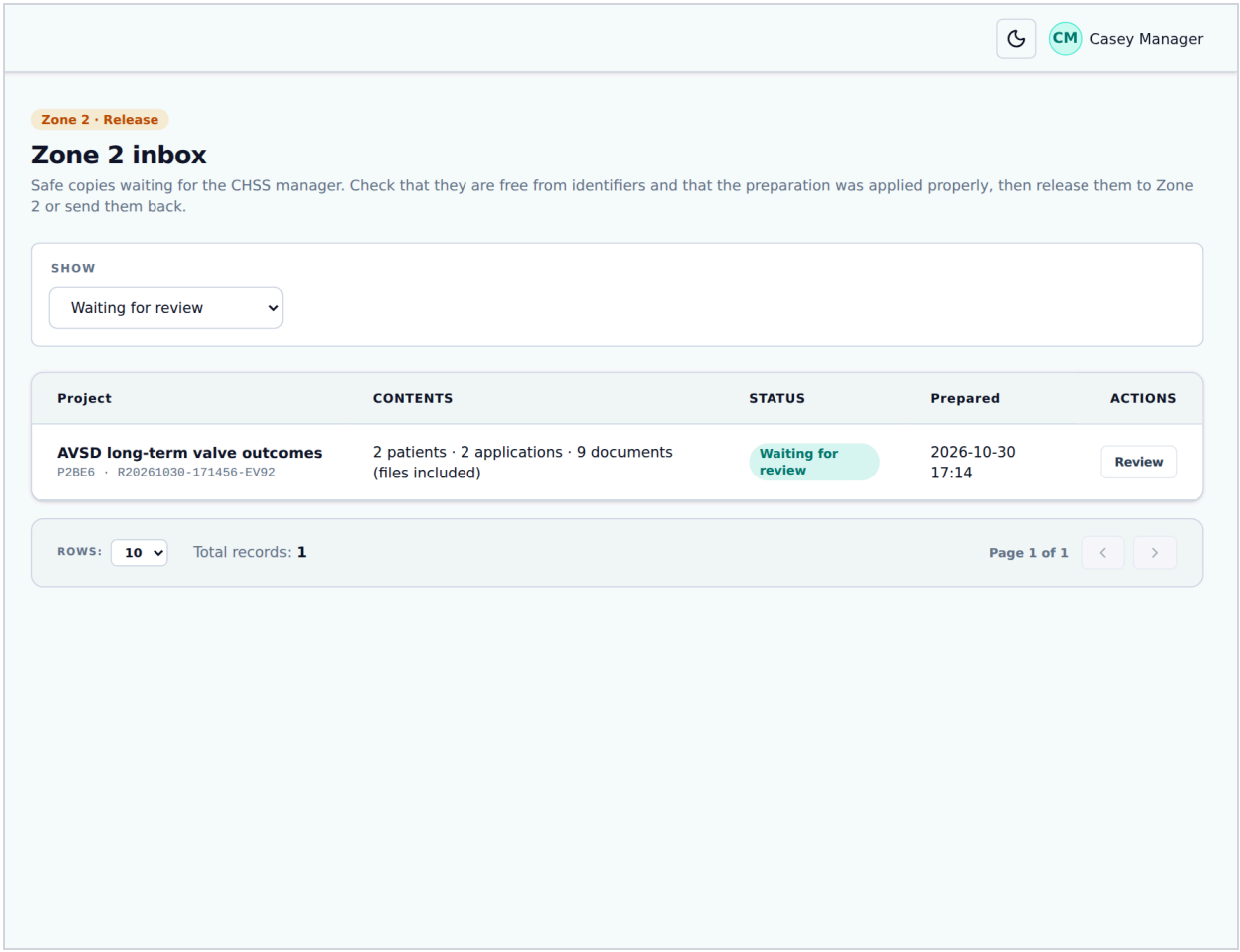


Figure 45. The Zone 2 inbox

1. Under **Show**, pick **Waiting for review**. Click **Review** on the release.
2. Check the tables: Hive shows the first rows of **patients.csv**, **applications.csv** and **documents.csv**. Look for anything that could identify someone.
3. If documents are included, open some of them. **Pick 5 at random** chooses a sample; **Open** shows a document. Every opening is logged.
4. If a document still identifies someone, click **Flag** and give the reason. The whole release is sent back and the document goes to Rejections to be redacted again by hand (chapter 5.5).
5. If something else is wrong, click **Send back** with the reason. The project returns to the analyst to fix and prepare again.
6. If everything is right, click **Release to Zone 2**. The safe copy is copied to the project's Zone 2 folder.

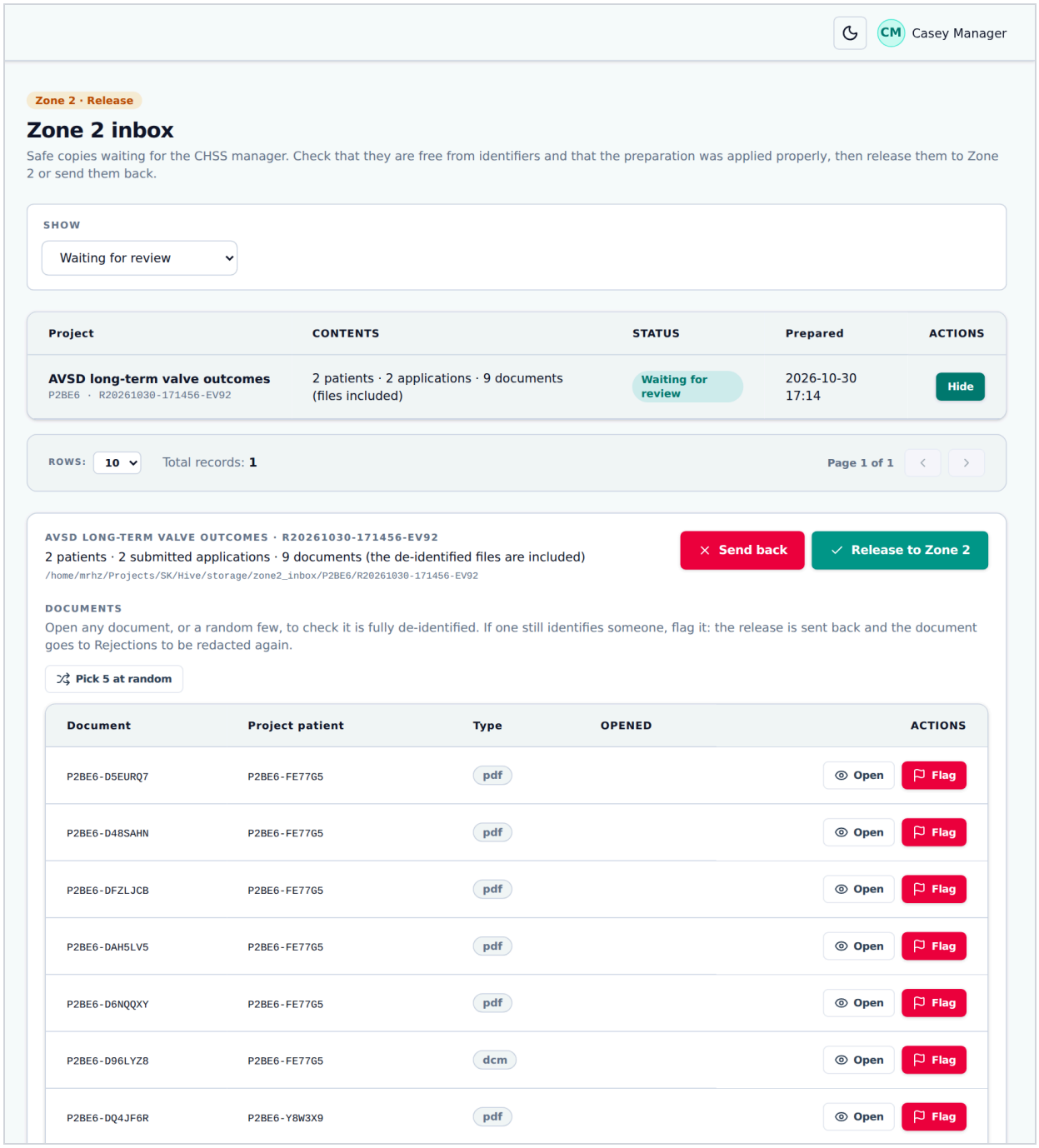


Figure 46. Reviewing a safe copy

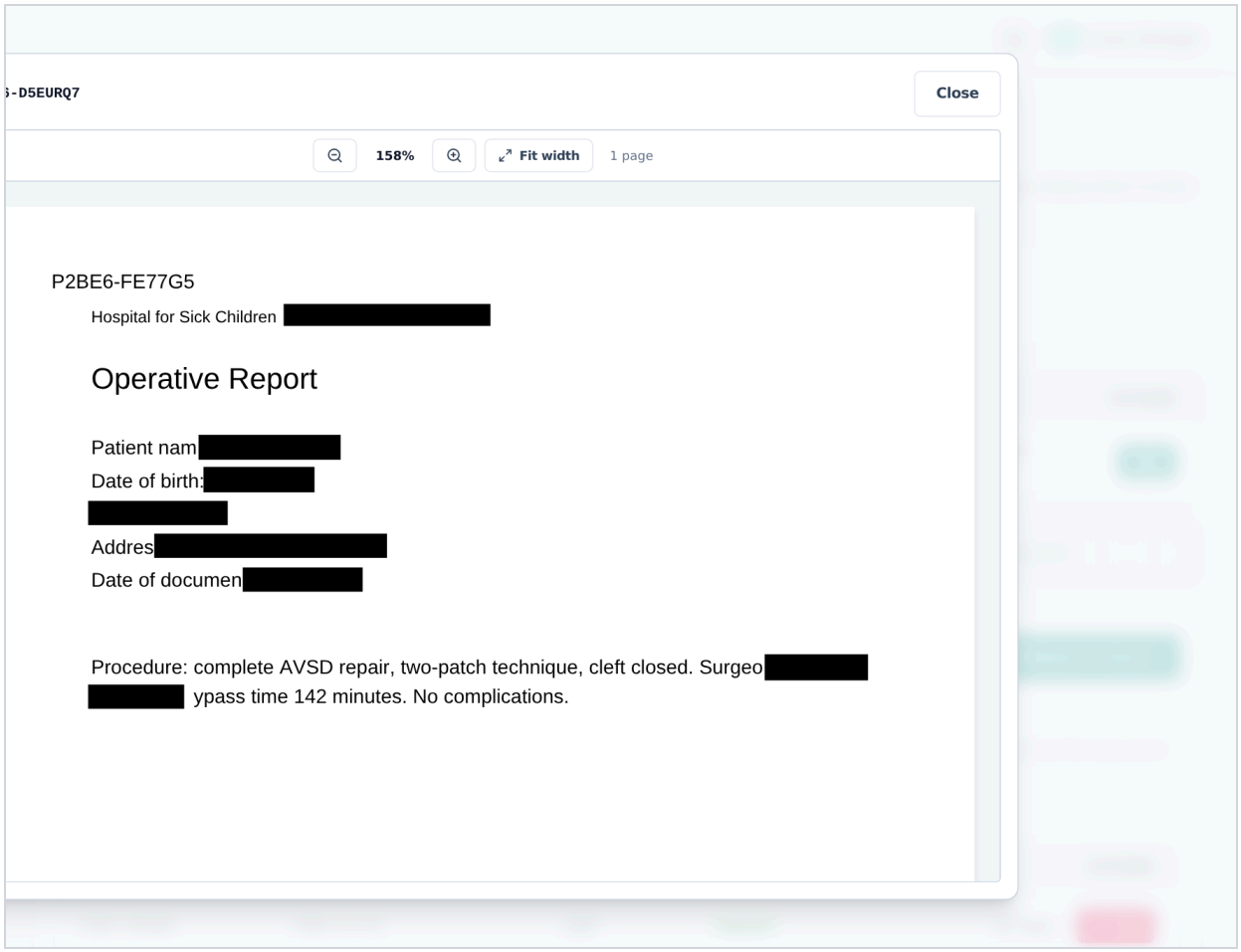


Figure 47. Spot-checking a released document: the name is the project ID

> **Protocol**
> §9.3.1: the Zone 2 inbox is an isolated area where the CHSS Manager or data manager checks that there are no identifiers and that the preparation was applied, before the data is released. §9.3.3: images and PDFs go to Zone 2 only with a demonstrated need and a full review.

7.2 Working with released data (researchers)

After release, project members find the data in the project's Zone 2 folder on Cloudera. In Hive, **Projects** shows the project with status **released**.

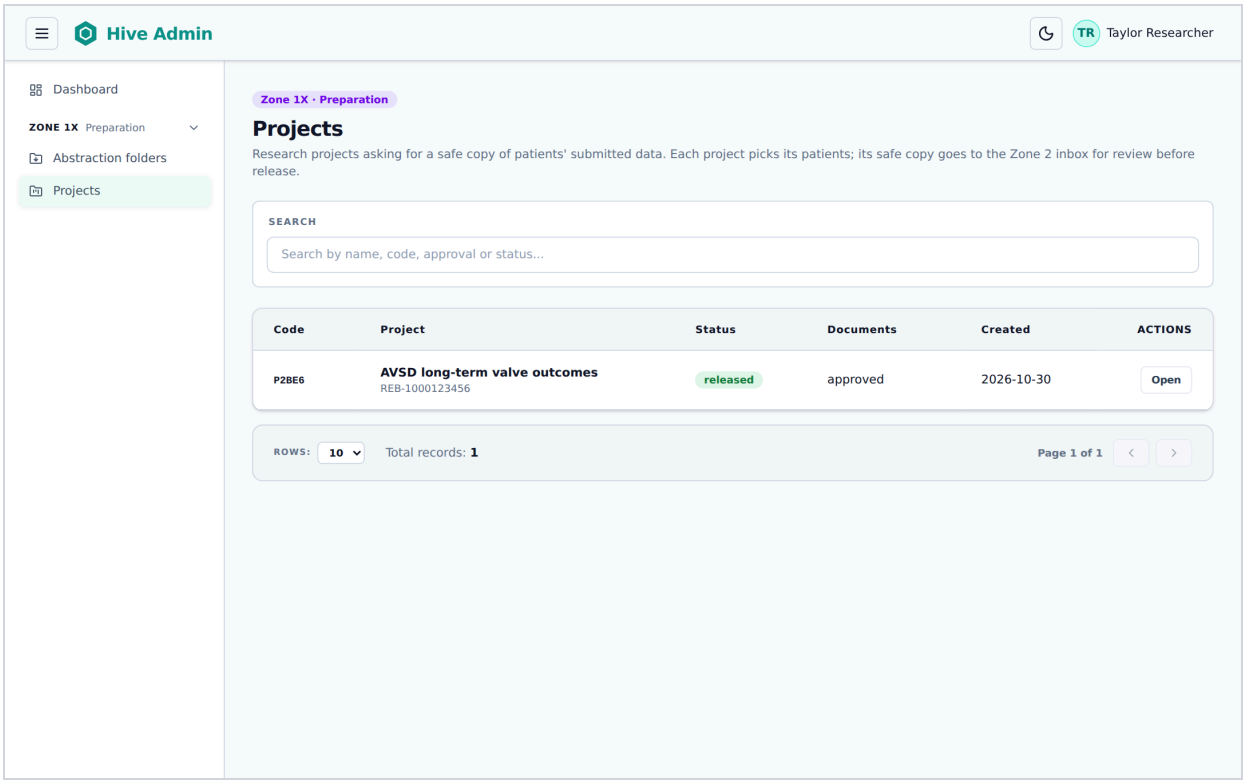


Figure 48. What a researcher sees: their released project

File	Contents
patients.csv	Project patient ID, age at registration, shifted dates, state and country
applications.csv	Project application IDs and shifted dates
documents.csv	Project document IDs, type, shifted date, size and file name
redcap_<dataset>.csv	Approved REDCap fields only (coded, number and date)
documents/	The de-identified files named by project ID (only if approved for documents)
data_dictionary.csv	The meaning, type and values of every column

Important

Never try to work out who a patient is, and never combine Zone 2 data with data from other sources to do so. Dates are shifted on purpose: intervals are true, calendar dates are not.

7.3 Analysis environment

Protocol

§9.3.2: approved users analyse data on standard analysis virtual machines (Anaconda, R, Python, GitLab, Office 365); custom machines (more memory, GPU, SAS Viya) on request.

§9.3.4: desktop SAS 9.4 linked to the project storage.

Not yet available in Hive

The analysis machines and SAS are provided by Cloudera and RIT, outside Hive. Today analysts use Cloudera sessions on the project’s Zone 2 folder.

[Picture to be added]

Picture of starting an analysis session on the project’s Zone 2 folder.

Figure 49. Opening the Zone 2 analysis environment

8. Zones 3 and 4

8.1 Zone 3: SharePoint

Protocol

§9.4: Zone 3 is the CHSS document repository on SharePoint, signed in with Entra ID. It must never hold PHI.

Not yet available in Hive

Zone 3 is outside Hive.

[Picture to be added]

Zone 3 SharePoint site

Figure 50. Zone 3 SharePoint site

8.2 Zone 4: REDCap and the patient portal

Protocol

§9.5.1: REDCap holds the consent and contact matrix; sites enter their own data through data access groups.

§9.5.2: one REDCap clinical dataset per active cohort.

§9.5.3: a standard periodic follow-up questionnaire (consents, transition at 18, health status, PedsQL, CHQ, ASQ, DASS21, SCORE15).

§9.5.4: the REDCap email client and survey tool send personalised links to patients (Figure 7).

Not yet available in Hive

The patient portal, the follow-up questionnaire and the survey invitations are in REDCap, outside Hive. Hive exchanges data with REDCap as described in chapters 5.11 and 5.12.

[Picture to be added]

Copy Figure 7 of the protocol here.

Figure 51. Patient follow-up workflow (protocol Figure 7)

9. Administration

Administration pages are for the CHSS Manager and the Application Specialist.

9.1 Users

The **Users** page lists everyone with a Hive account. Everyone must already be able to sign in to Cloudera.

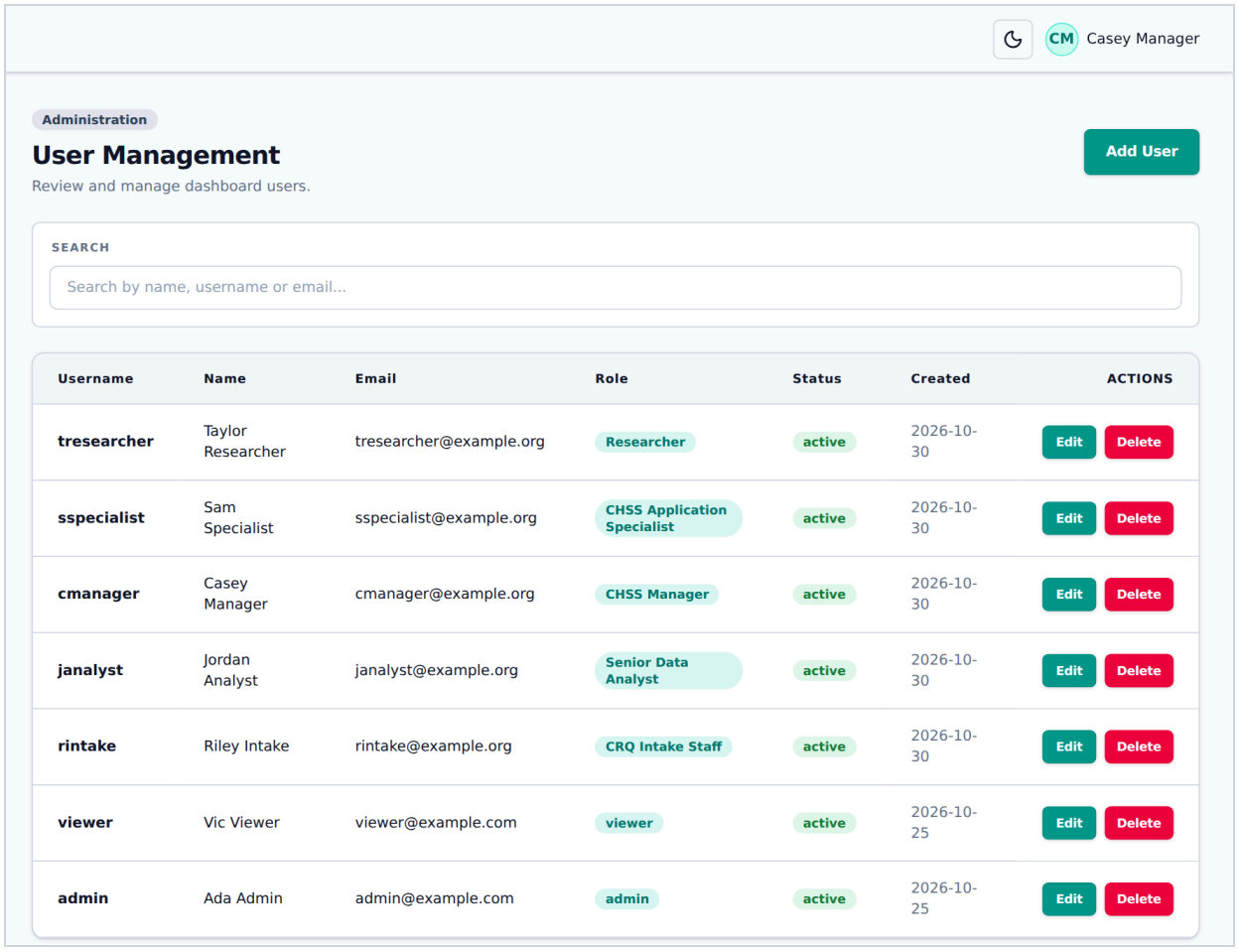


Figure 52. User Management

1. Click **Add User**.
2. Enter the **Username** exactly as Cloudera knows it, the email, first and last name.
3. Pick the **Role** and leave **Active** ticked.
4. Click **Create user**.

To change someone's role or take away their access, click **Edit** and change the **Role**, or untick **Active**. Inactive and unknown accounts are refused and the attempt is logged in the Access log.

> **Protocol**
> §6.1: role-based access with least privilege. The CHSS Manager reviews access requests, documents compliance and reviews the user directory.

9.2 Roles

The **Roles** page lists the roles with the permissions each one has. Click **Add Role** for a new one, or **Edit** to change one. Permissions are grouped by zone; tick what the role may do. Changes take effect immediately.

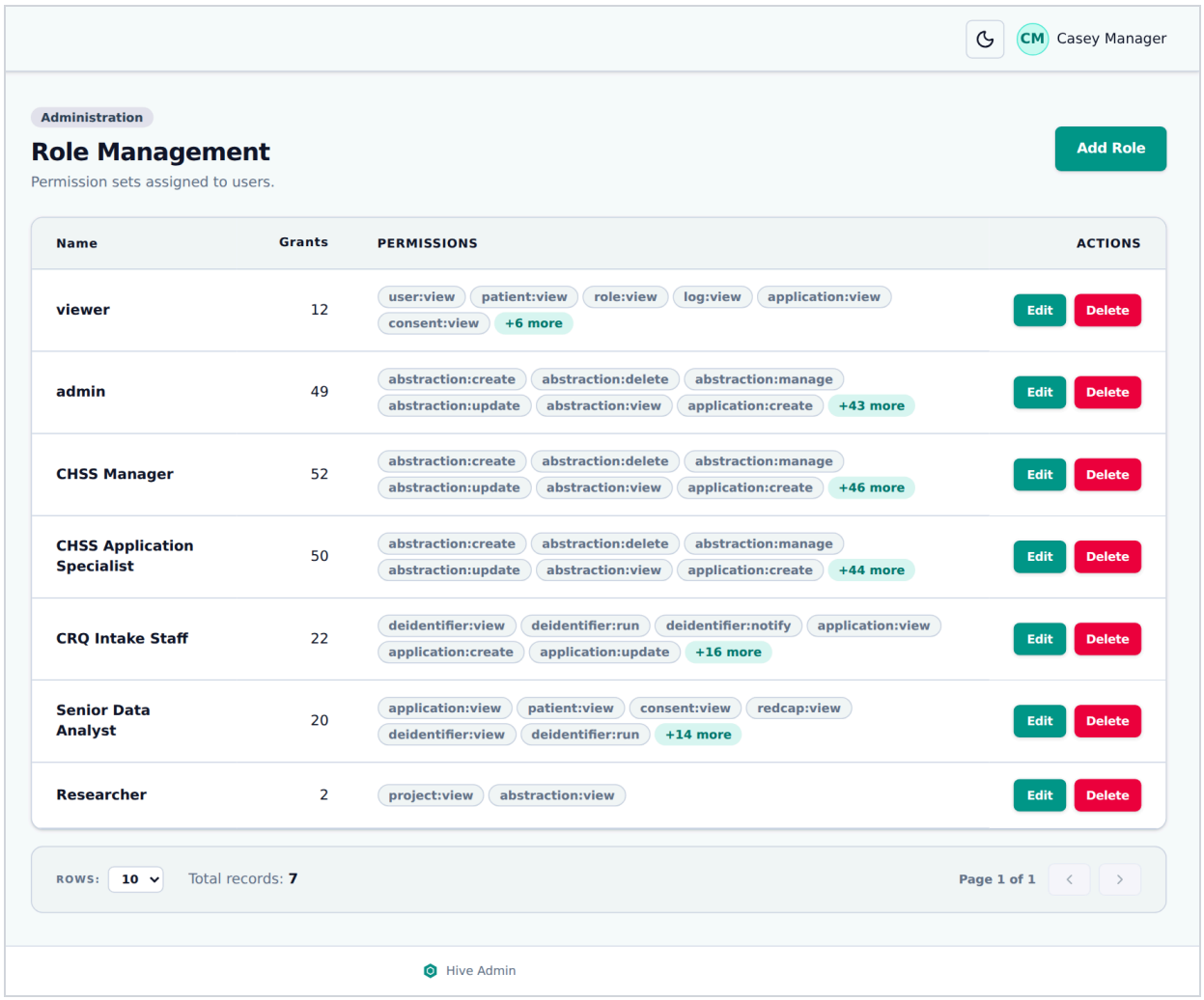


Figure 53. Role Management

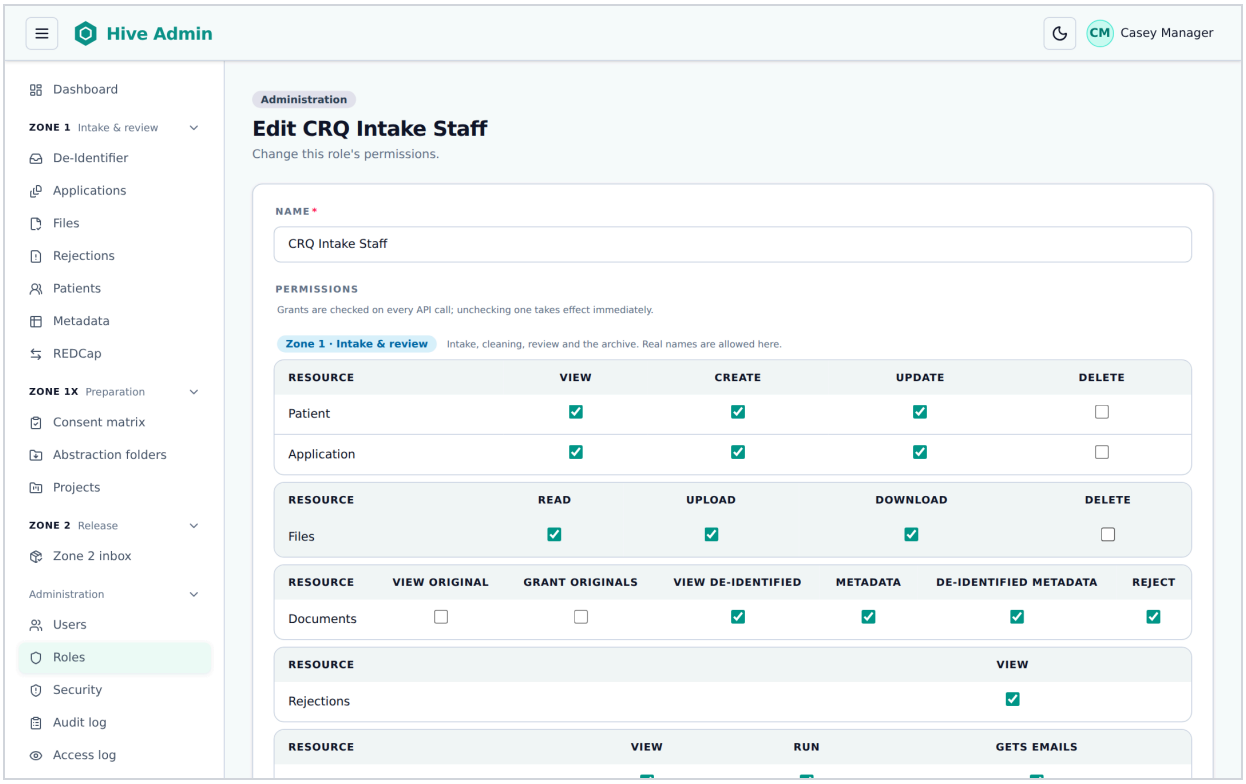


Figure 54. Editing a role: permissions grouped by zone

The full list of permissions is in chapter 10.4.

9.3 Security

The **Security** page is used during a security incident.

Locking Hive

1. Under **Lock the app**, write the **Reason** and pick the **Level**.
2. Click **Lock now**. Everyone without security management is stopped at once and sees the red banner. The CHSS Manager, Application Specialist and Host IT are emailed.
3. To unlock, click **Unlock** in the red banner and give the reason. The lock and unlock are recorded in the Audit log and emailed.

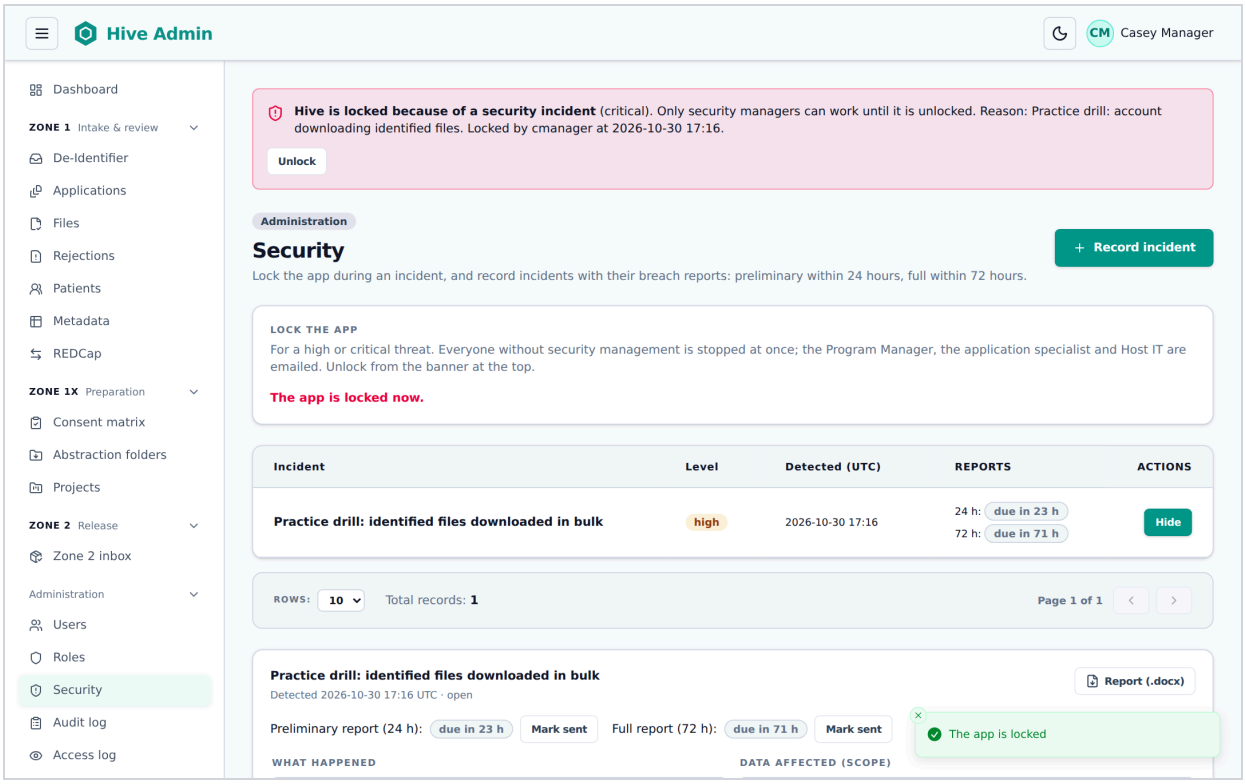


Figure 55. Hive locked: the manager can still work and unlock

Hive also watches for unusual activity every hour (many downloads of identified files, large exports, refused requests, failed sign-ins). Low alerts are emailed to the Application Specialist; high and critical alerts to the CHSS Manager, Application Specialist and Host IT. At the critical level Hive locks itself.

Recording an incident and the breach reports

1. Click **Record incident**. Enter the **Title**, **Level**, **Detected at** (the 24 and 72 hours count from here) and **What happened**. Click **Record incident**.
2. Open the incident and fill in **Data affected (scope)**, **Risk assessment**, **Containment actions** and **Affected parties**. Click **Save**.
3. Click **Report (.docx)** to download the breach report. Send the preliminary report within 24 hours and click **Mark sent**; send the full report within 72 hours and click **Mark sent**.
4. When everything is done, click **Close incident**.

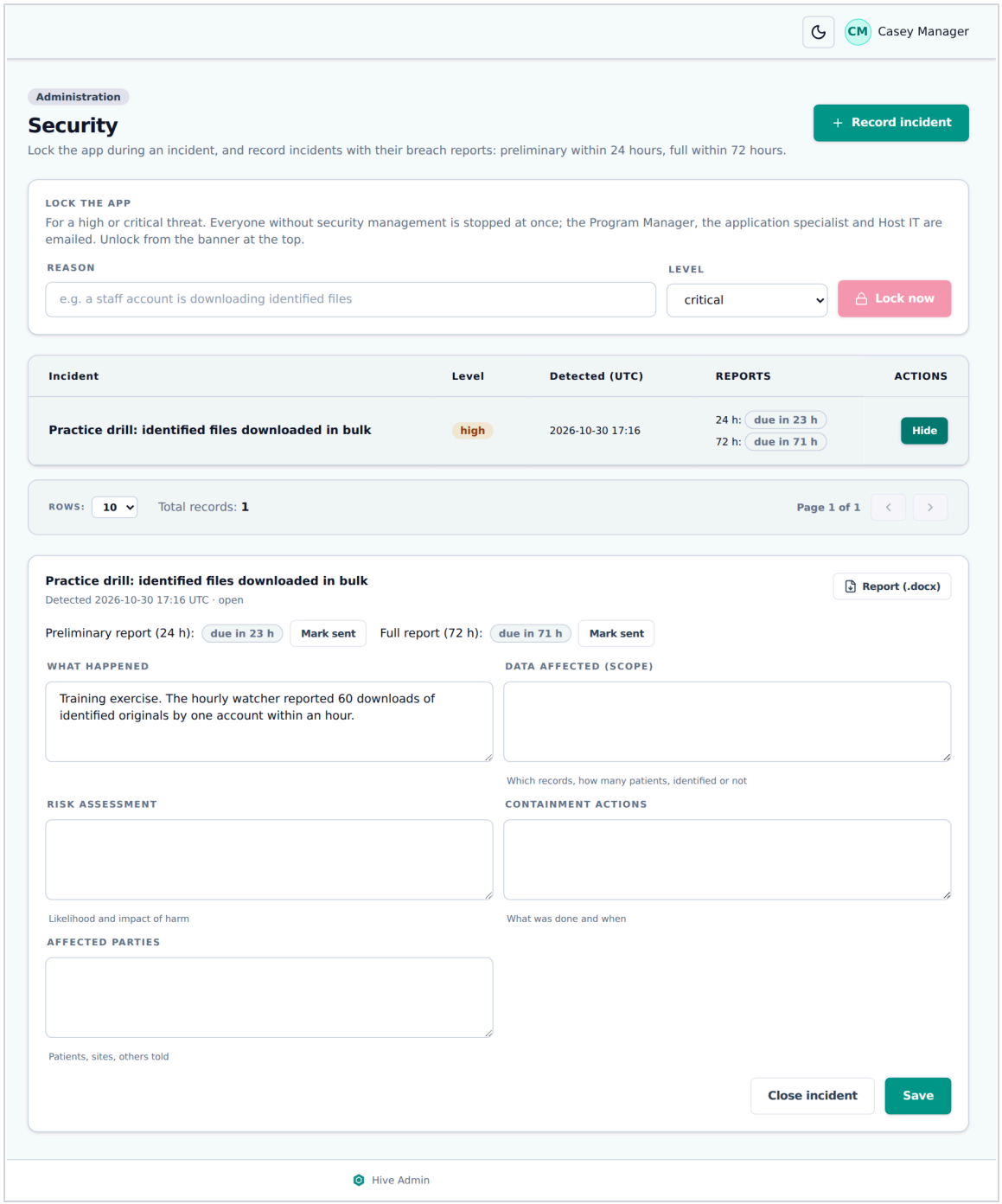


Figure 56. An incident with its 24-hour and 72-hour report deadlines

Protocol

§6.5: a breach report includes the time of detection, scope, risk assessment, containment and affected parties. The preliminary report is due within 24 hours and the full report within 72 hours.

> **Not yet available in Hive**
>
> §6.5: incidents are created automatically when a breach is detected. Today the hourly watcher emails and can lock Hive, but a person records the incident.

9.4 Audit log

The **Audit log** records every change: who did what, to which record, and when. It cannot be edited or deleted. Filter by **Entity type**, **Record id**, **Who**, **Action**, **From** and **To**. Click a row to see the values before and after.

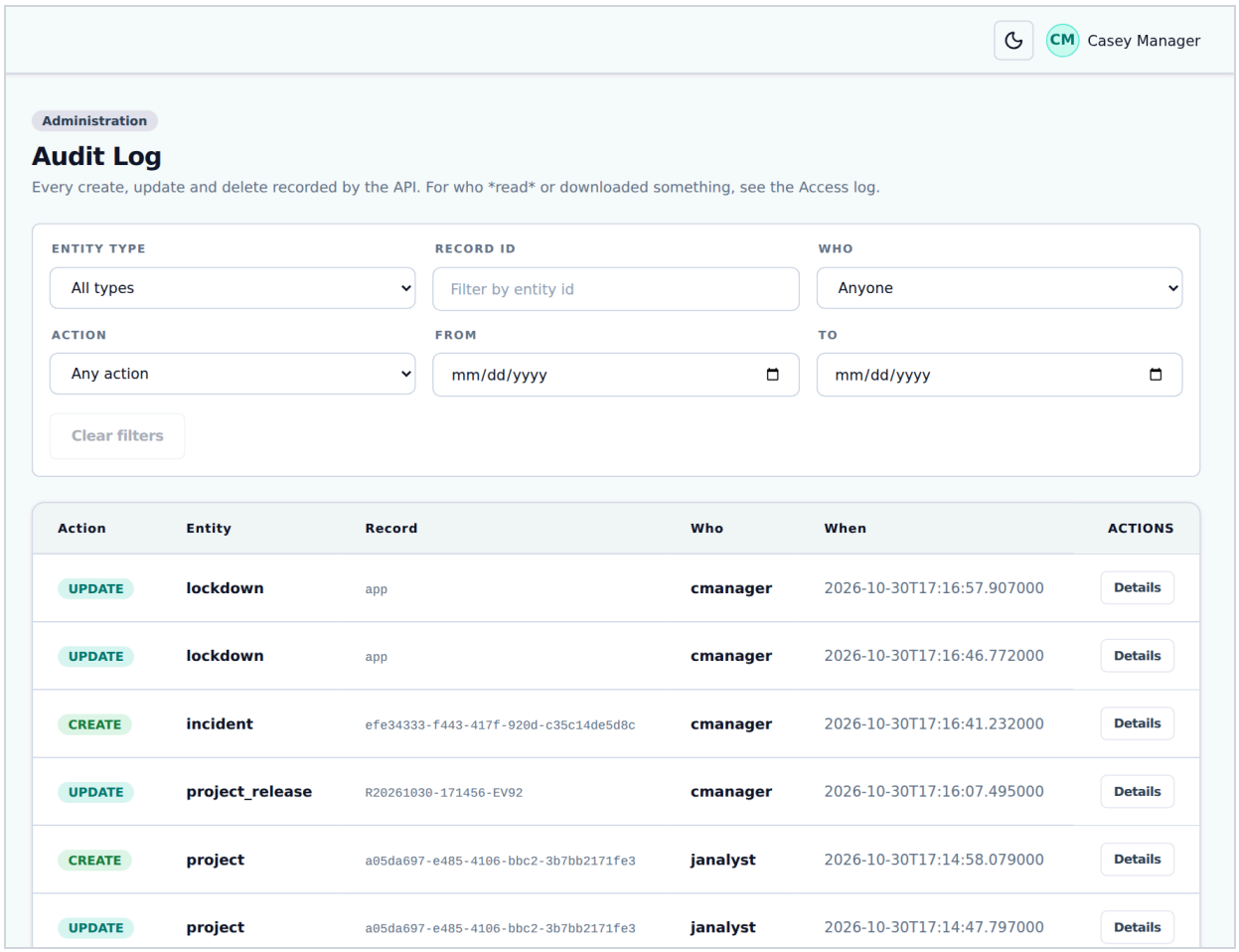


Figure 57. The Audit log

9.5 Access log

The **Access log** records who read, downloaded or exported what, and who was refused. **Identified** shows whether an original (with PHI) was opened. Tick **Disclosures only** to see only reads where identified PHI left Hive.

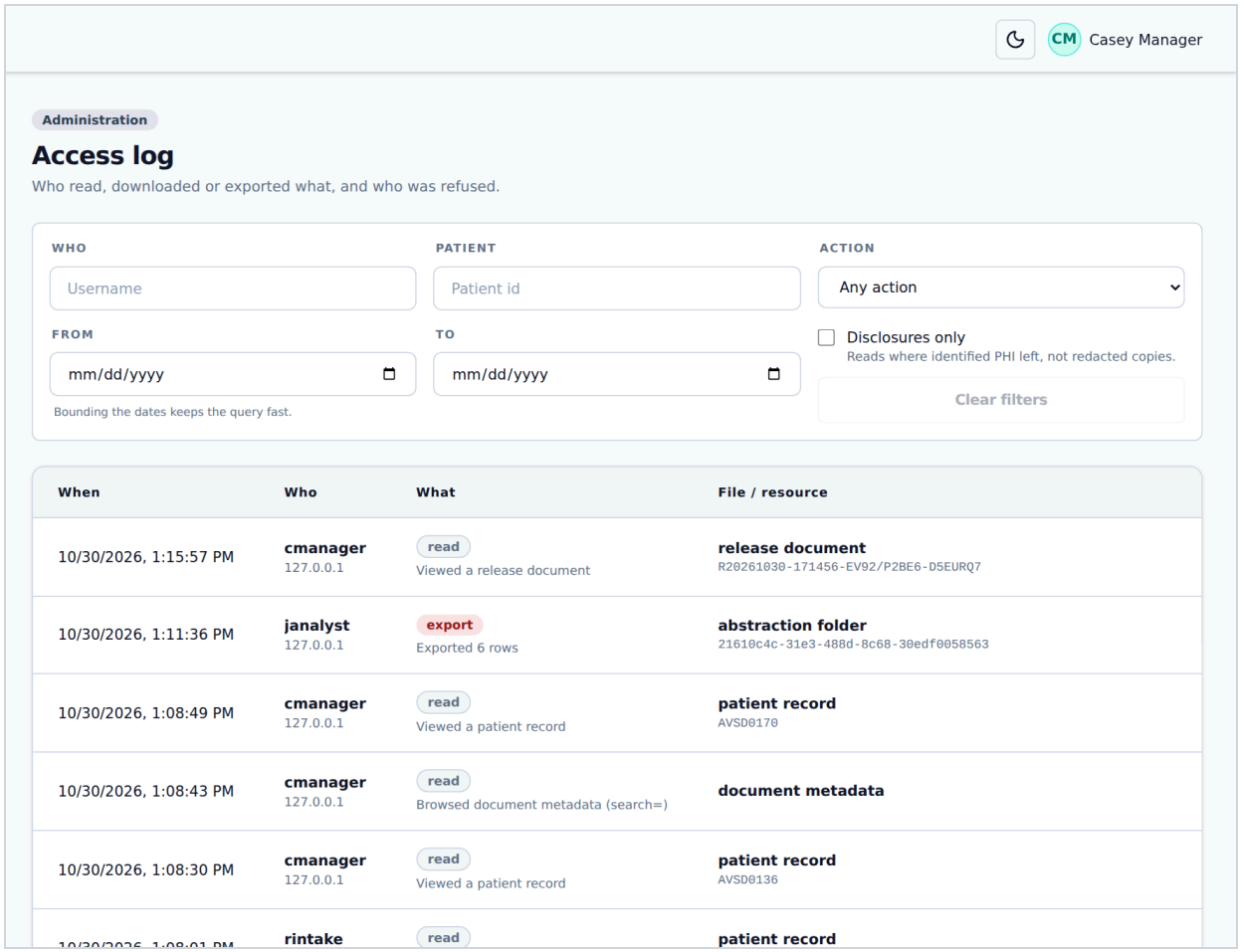


Figure 58. The Access log

Protocol

§6.5–6.6: user activity, data access and changes are logged with user, action, record and time; unauthorised attempts are logged; the log cannot be changed. Data movements and transformations are traceable (lineage).

10. Reference

10.1 Retention and time limits

What	Time limit	Section
Abstraction folder	Deleted after 90 days (3 months) unless extended	§9.1.6
Project workspace (Zone 1X)	Deleted 90 days after first built unless extended with approval	§9.2.2
Zone 2 access	Limited to one project; reviewed every 6 months; removed when the project ends	§5
Breach report	Preliminary within 24 hours, full within 72 hours of detection	§6.5
Date shift in Zone 2	Random per patient, −1826 to +1826 days	§9.2.4

10.2 Document types

Code	Document	Found from words in the file name
ENROLLMENT	Enrolment form	enrollment, enrolment, registration form
CONTACT	Contact form	contact form, change of contact, contact
CONSENT	Consent form	consent, assent
OPERATIVE	Operative report	operative, op note, surgical report
DISCHARGE	Discharge summary	discharge, dc summary
ECHO_REPORT	Echocardiogram report	echo, echocardiogram, TTE, TEE
ANGIO_REPORT	Angiography report	angio, angiogram, cath
MRI_REPORT	MRI report	MRI, CMR, cardiac MRI
ECG_REPORT	ECG report	ECG, EKG
IMAGE	Image	JPG, PNG and other pictures
STS_EXTRACT	STS extract	Excel or CSV with “sts”
OTHER / UNKNOWN	Anything else / not found	Set by hand with Set type & date

10.3 Where files are kept

Stage	Storage
Waiting for de-identification	Intake folder (inbox)
De-identified, application still a draft	Pending storage: original/ and de_identified/ under the patient code
Failed or excluded	Pending storage: failed/ and needs_attention/
Application submitted	Archive (originals) and active storage (de-identified copies)
Abstraction folder	Abstraction storage, one folder per user
Project workspace	Project storage: <project code>/workspace/
Release archive	Project storage: original data, id_map.csv, the safe copy
Waiting for review	Zone 2 inbox
Released	Zone 2: <project code>/<release>/

10.4 Permissions

Permission	Allows
patient: view / create / update / delete	See, add, change and delete patients
application: view / create / update / delete	See and review applications; delete is needed to reject as “not eligible”
files: read / upload / download / delete	Files page, adding, downloading and deleting documents
files: view original	Open originals (before submission, or with a grant after)
files: grant originals	Give access to an archived original (CHSS Manager only)
files: view de-identified / metadata / de-identified metadata	Open redacted copies and the metadata of either copy
files: reject	Reject documents, mark leaded images
rejection: view	Rejections page
deidentifier: view / run / gets emails	De-Identifier page, start runs and Copy from Carbon, receive intake emails
redcap: view / import / export	REDCap page, imports, follow-up and code lists
consent: view / update	Consent matrix, record and import decisions
project: view / create / update / delete	Projects you are a member of

Permission	Allows
project: workspace storage	Build, extend and delete workspaces
project: all projects + members	See every project, add and remove members
abstraction: view / create / update / delete / everyone’s folders	Your own folders; everyone’s folders with the last one
zone2: prepare	Prepare a safe copy
zone2: review	Zone 2 inbox: release or send back (CHSS Manager)
user / role: view / create / update / delete	Users and Roles pages
log: view	Audit log and Access log
security: view / manage	Security page; lock and unlock, record and close incidents

10.5 Common questions

Question	Answer
A page I need is not in my sidebar.	Your role does not include it. Ask the CHSS Manager to review your role.
A file was excluded with “no patient code”.	The code is missing or not in the accepted format in both the folder and the file name. Ask the site to resend it with the code, then Retry.
Submit application stays grey.	Confirm the patient’s details in step 1, and make sure no document is undecided or rejected. Hover over the button to see why.
I can no longer open an original.	After submission, originals need a grant from the CHSS Manager (chapter 5.6).
Prepare safe copy is refused.	Read the message: an approval reference, consent confirmed and at least one patient are needed, and patients kept out by the consent matrix must be removed from the cohort.
The red “Hive is locked” banner is showing.	A security incident is being handled. Wait until it is unlocked.
I see a real identifier in a de-identified copy.	Reject the document with the reason (chapter 5.4). Never copy or share it.

10.6 Glossary

Term	Meaning
Application	One patient’s submission: their details and documents
CRQ	The team that receives and checks the data in Zone 1
De-identified copy	The redacted copy of a document, named with _R or _RR
DICOM	The standard format for medical images
Leaded image	An image with identifiers in lead letters; kept out of Zone 2
PHI	Personal health information that can identify a patient
Project ID	A patient’s ID in one project only; cannot be matched to the CHSS code
Safe copy	The de-identified, date-shifted data prepared for Zone 2
UBRS	The patient code used instead of the patient’s identity
Workspace	The project’s Zone 1X folder for data cleanup